Rheinisch–Westfälische Technische Hochschule Aachen

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A Machine Learning Model for the Prediction of Hessian Matrices with atomistic based quantities

Constructing and Testing a Machine Learning Model on a Small Data Set with Systems of Two to Four Atomic Molecules for Exploratory Investigations

Research Internship Report

submitted by

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List of Abbreviations

APF	Atom Pair Focused
BO	Born–Oppenheimer
Case	Case–Specific
CART	Classification and Regression Tree
CS	Coordinate System
DFT	Density Functional Theory
DFTB	Density Functional Tight Binding
ETR	Extra Tree Regression
ES	Electrostatic
Gen	General
HVF	Harmonic Vibrational Frequencies
Idx	Indexed
KS–DFT	Kohn–Sham Density Functional Theory
LDA	Local Density Approximation
MI	Main Inertia
ML	Machine Learning
Perm	Permuted
PES	Potential Energy Surface
RFR	Random Forest Regression
RMSD	Root–Mean–Squared–Distance
XC	Exchange and Correlation

1 Motivation

The interest of automatization in chemistry has risen in the last decade. There are approaches for the use of robotics to automate experiments.[1,2] And also in theoretical chemistry, approaches are developed to automatize workflows of computations.[3] These processes lead to an increase of useable data. Furthermore, databases are published like the benchmark set GMTKN55 (**G**eneral **M**aingroup **T**hermochemistry **K**inetic & **N**oncovalent interactions)[4] or the CCCBDB (**C**omputation **C**hemistry **C**omparison and **B**enchmark **D**ata**b**ase)[5]. This increase of available data and of automatization are both of interest for machine learning (ML) approaches, as those are based on such data.

ML models always consist of features and a target. In the subclass of supervised ML the model is trained with features and target. Afterwards the trained model is used to predict the target only by the information of the features. To build an efficient model, the features should be easily accessible, while the target should be computationally costly. Thus, the ML model has the advantage to accelerate the process of determining the target.

The harmonic vibrational frequencies (HVF) are valuable quantities in quantum chemistry. They are used for the computation of infrared[6] and Raman spectra[7], and also for the calculation of thermodynamic quantities via a quantum mechanical description of the nuclear degree of freedom[8]. Furthermore, they are necessary for the transition state search, as the imaginary HVF indicate transition states.[9] The HVF are computed from the eigenvalues of the Hessian matrix. In chemistry, a Hessian is a matrix of second derivatives of the energy E with respect to the $3M$ coordinates q_i of the M atoms in a molecular system.[10] This Hessian is depicted in Eq. (1.0.1).

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 E}{\partial x_A \partial x_A} & \cdot\cdot & \frac{\partial^2 E}{\partial x_A \partial y_M} & \frac{\partial^2 E}{\partial x_A \partial z_M} \\ \frac{\partial^2 E}{\partial x_A \partial y_A} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 E}{\partial z_M \partial x_A} & \cdot\cdot & & \frac{\partial^2 E}{\partial z_M \partial z_M} \end{pmatrix} \tag{1.0.1}$$

The computation of the Hessian is costly.[11] Thus, the acceleration of the computation is of great interest as a target for ML.

There are already approaches for accelerating the computation of the Hessian in literature.[11–13] Two of these will be breifly discussed in the following.

Spicher and Grimme developed an approach to compute the Hessian with a low–level methods at a higher level geometry. A bias depending on the root–mean–squared–distance (RMSD) is added to the potential energy surface (PES) of the low–level method, aiming to minimize the energy at the geometry of the high–level method. On this new PES the Hessian matrix can be computed, as the harmonic oscillator–approximation is applicable at the higher–level geometry.[11] The contributions of the bias potential energy is Afterwards detached from the eigenvalues.

Sanders et al. proposed another approach, which is based on the assumption, that the Hessian matrix is a sparse matrix. They use a matrix, that describes the sparse matrix in an incoherent basis. The sparse matrix is approximated by using the matrix in the incoherent basis partially. Hence, instead of computing the full matrix only a part of it needs to be computed. This can be either used to accelerate the computation or to increase the precision by using high–level methods.[13]

In this research a new approach shall be investigated. As noted above, due to automatization and large available data sets the use of a ML model might be an efficient method to predict the costly Hessians. Here, a supervised ML approach was chosen, where the target to predicted are the Hessian matrix elements. The features for the ML model are based on computational efficient atomistic–based quantities, obtained from a single–point electronic structure calculation at a semiempirical tight–binding level.dd

2 Theory

Four topics will be discussed in the theory as this research relies on those. At first, the Euler angles, as these are used for the transformation of the coordinate system to achieve a specific coordinate frame. Secondly, the GFN2–xTB method, as it is used to compute the features and target. Thirdly, the Hessian matrix, as its elements are the target of the machine learning model. At last, the used ML algorithms are described for further discussion in the results.

2.1 Euler Angles

One method of describing coordinate system (CS) transformations, are Euler angles. They serve as a mathematical tool to connect one CS with another and will be explained in the following with an example.[14]

The initial CS $\mathbf{x}, \mathbf{y}, \mathbf{z}$, shown in Fig. 2.1 can be transformed into the $\mathbf{x}', \mathbf{y}', \mathbf{z}'$ CS by three rotations. The distance between the vectors $\mathbf{L}_\mathrm{a}$ and $\mathbf{L}_\mathrm{b}$ is shown in red, and is the intersection $\mathbf{L}$ between the $\mathbf{xy}$ and $\mathbf{x'y'}$ planes. The three Euler angles α, β and γ are defined as follows: α is the angle between $\mathbf{x}$ and $\mathbf{L}_\mathrm{a}$, β is the angle between the two z–axes $\mathbf{z}$ and $\mathbf{z}'$, and γ is the angle between $\mathbf{L}_\mathrm{a}$ and $\mathbf{x}'$. These angles are then used to define rotation matrices for the CS transformation.

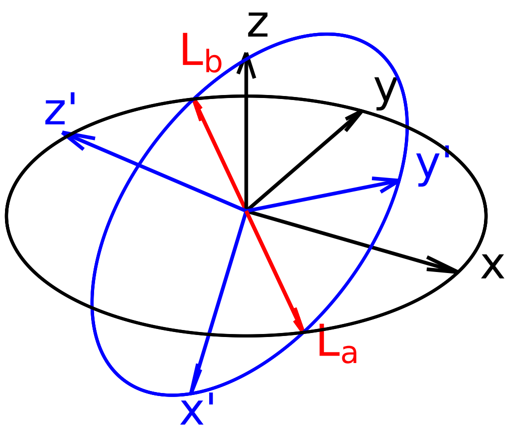


Figure 2.1: Depiction of two coordinate systems $\mathbf{x}, \mathbf{y}, \mathbf{z}$ (black) and $\mathbf{x}', \mathbf{y}', \mathbf{z}'$ (blue). The red intersection of the **xy** and **x'y'** planes is the euclidean distance between the red vectors $\mathbf{L}_a$ and $\mathbf{L}_b$.

For the rotation of the CS it is necessary to apply the rotation matrices in a specific order. The first rotation is around **z** by the angle α, followed by the rotation around **x** by β. Then a rotation around the **z** axis is done by the angle γ.
This is mathematically described with the Givens–rotation matrices $\mathbf{R}_z(\theta)$, and $\mathbf{R}_x(\theta)$ shown in Eq. (2.1.1) and Eq. (2.1.2), respectively. These are the rotation matrices for the Givens–rotation around the **z** and **x** axes. Where θ is an arbitrary angle.

$$\mathbf{R}_z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{2.1.1}$$

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} \tag{2.1.2}$$

The full rotation of the coordinate system in dependency of the Euler angles is shown in Eq. (2.1.3).

$$\mathbf{R} = \mathbf{R}_z(\gamma) \times \mathbf{R}_x(\beta) \times \mathbf{R}_z(\alpha) \tag{2.1.3}$$

2.2 Electronic Structure Theory

In this work, the features and target for the ML model are derived by the use of the GFN2–xTB method. Therefore, a short introduction to the semiempirical method GFN2–xTB is given. Before that, the mean–field theories Hartee–Fock and density functional theory (DFT) are discussed. DFT builds the theoretical basis for density functional tight–binding methods (DFTB), like GFN2–xTB. Atomic units are used for all equations in this section. This introduction of Hartree Fock and density functional theory is based on reference [10].

2.2.1 Hartree–Fock Model

Molecular systems in quantum mechanics can be described by a many–body wave function eigenvalue problem. The solution of this many–body wave function eigenvalue problem is given by solving the Schrödinger equation. The separation of its time–dependency, neglecting relativistic effects and the introduction of the Born–Oppenheimer (BO) approximation results in the electronic Schrödinger equation (Eq. (2.2.1)).

$$\hat{H}_{\mathrm{e}}\Psi_{\mathrm{e}} = E_{\mathrm{e}}\Psi_{\mathrm{e}} \tag{2.2.1}$$

The solution of this eigenvalue equation leads to the electronic energy E_{e}. In the Hartee–Fock model, the electronic many–body–wave function Ψ_{e} is described by a single Slater determinant (Eq. (2.2.2)). The Pauli principle introduces the condition of the antisymmetry of the wave function. This condition is fulfilled by Slater determinants. The Slater determinant consists of k spin–orbital wave functions ϕ_i, and every wave function is permuted with the coordinates $\mathbf{r}_i$ of the N electrons. Furthermore, the wave function is normalized by the factor $(N!)^{-\frac{1}{2}}$.

$$\Psi_{\mathrm{e}} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{r}_1) & \phi_2(\mathbf{r}_1) & .. & \phi_k(\mathbf{r}_1) \\ \phi_1(\mathbf{r}_2) & \phi_2(\mathbf{r}_2) & & \vdots \\ \vdots & & \ddots & \vdots \\ \phi_1(\mathbf{r}_N) & \dots & .. & \phi_k(\mathbf{r}_N) \end{vmatrix} \tag{2.2.2}$$

The electronic Hamiltonian $\hat{H}_e$ is shown in Eq. (2.2.3) and consists of the kinetic energy of the electrons and the potential energy between electrons and nuclei as also between the electrons. Furthermore, the nuclear repulsion energy, which is approximated as a constant, is added. The kinetic energy of the nuclei is separated, due to the BO–approximation.

$$\hat{H}_e = \sum_{i=1}^{N}\left(-\frac{1}{2}\nabla_i^2 - \sum_{A=1}^{M}\frac{Z_A}{|\mathbf{R}_A - \mathbf{r}_i|}\right) + \sum_{i=1}^{N}\sum_{i>j}^{N}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|} \tag{2.2.3}$$

Here, $\mathbf{R}_A$ are the coordinates of atom A, $\mathbf{r}_i$ the coordinates of electron i and Z_A the nuclear charge. While, ∇_i^2 is the second derivative with respect to every coordinate of electron i.
The electronic energy can now be described as an expectation value of a Slater determinant and Hamiltonian in Eq. (2.2.4). This leads to the description of the energy by the one–electron operator $\hat{h}_i$, the coulomb operator $\hat{J}_i$ and the exchange operator $\hat{K}_i$.

$$E_e = \sum_{i=1}^{N}\langle\phi_i|\hat{h}_i|\phi_i\rangle + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\Big(\langle\phi_j|\hat{J}_i|\phi_j\rangle - \langle\phi_j|\hat{K}_i|\phi_j\rangle\Big) \tag{2.2.4}$$

The Rayleigh–Ritz variational principle (s. Eq. (2.2.5)) states, that the true total energy E_0 of the system is always beneath or equal to the energy corresponding to the approximated wave function here, the Slater determinant. This allows the search of the best possible energy by optimizing the wave function (or its parameters).

$$E_0 \leq \frac{\langle\Psi|\hat{H}|\Psi\rangle}{\langle\Psi^*|\Psi\rangle} \tag{2.2.5}$$

The determination of the total energy of the system, described by a single Slater determinant, is an eigenvalue problem and can be solved by minimizing the energy. A constrained optimization using the method of Lagrange multiplier is employed, under the condition of orthonormal spin orbitals ϕ_i. Hence, the inner product of the wave function $\langle\phi_i|\phi_j\rangle = \delta_{ij}$ corresponds to the Kronecker delta (Eq. (2.2.6)).

$$\langle\phi_i|\phi_j\rangle = \delta_{ij} = \begin{cases} 0 \text{ if } & i \neq j \\ 1 \text{ if } & i = j \end{cases} \tag{2.2.6}$$

The derivative of the Lagrangian functional leads to the canonical Hartree–Fock equation (2.2.7). Solving these yields orbitals, that minimize the energy of the Slater determinant.

$$\hat{f}(\mathbf{r}_i)\phi_k(\mathbf{r}_i) = \epsilon(\mathbf{r}_i)\phi_k(\mathbf{r}_i) \tag{2.2.7}$$

$$\hat{f}(\mathbf{r}_i) = \hat{h} + \sum_{j=1}^{N}(2\hat{J}_j - \hat{K}_j) \tag{2.2.8}$$

The Fock–operator $\hat{f}(\mathbf{r}_i)$ is an effective one–electron operator.
Solving the Hartree–Fock equations is equal to determining the spin–orbitals in an iterative process, as the Fock–operator itself depends on the spin–orbitals. Important to point out is that the Fock–operator consists of the one–electron operator, the exchange–operator and the coulomb operator.
The spatial part of the spin–orbitals are described via a basis set approximation. This LCAO–approach is shown in Eq. (2.2.9), where the spin–orbital wave–functions are the sum of K basis functions χ_μ with a linear combination coefficient c_μ.

$$\phi_i(\mathbf{r}) = \sum_{\mu}^{K} c_\mu \chi_\mu(\mathbf{r}) \tag{2.2.9}$$

Merging the basis–set approach into Eq. (2.2.7) leads to the Roothaan–Hall equation shown in Eq. (2.2.10). Here, $\mathbf{F}$ is the Fock matrix, $\mathbf{C}$ a matrix of coefficients c_μ, $\mathbf{S}$ the overlap matrix between the atomic basis functions and ϵ the diagonal eigenvalue matrix. This equation is used to iteratively compute a solution for the energy. Starting with an initial guess of the Fock matrix, the matrix $\mathbf{C}$ is computed. With $\mathbf{C}$ the density matrix $\mathbf{D}$ is calculated. The density matrix $\mathbf{D}$, is then used for the calculation of the new Fock matrix, etc.

$$\mathbf{FC} = \mathbf{SC}\epsilon \tag{2.2.10}$$

2.2.2 Kohn–Sham Density Functional Theory

As mentioned above, the correlation between electrons is not described in the Hartree–Fock model, due to the use of a single Slater determinant.

The correlation energy can be accounted for by two different approaches. Instead of using one single Slater determinant, multiple combinations of Slater determinants can be used. This leads to post–Hartree–Fock methods like full configuration interaction or MP2. The second approach is density functional theory (DFT).

The Hohenberg–Kohn theorems build the basis for DFT. These two theorems postulate, that the ground state energy is a functional of the electron density, and that this functional follows the variational principle.

If a single determinant is used (as in KS–DFT, see below), the energy can be described via the density of the electrons, which is defined in Eq. (2.2.11). Where, n_{occ} is the occupation number of the spin–orbital.

$$\rho(\mathbf{r}) = \sum_{i=1}^{n_{\text{occ},i}} n_{\text{occ},i} |\phi_i(\mathbf{r})|^2 \tag{2.2.11}$$

The general approach is to describe the energy in dependence of the electron density $E[\rho]$. There are different approaches for this connection. The most common approach, Kohn–Sham Density Functional Theory (KS–DFT) reintroduces orbitals to compute the kinetic energy. The kinetic energy is inexact, due to the assumption of non–interacting electrons. The energy (s. Eq.(2.2.12)) in KS–DFT consists of the inexact kinetic energy, coulomb integral, the exchange correlation energy and the potential field. The energy E_{XC} consists mainly of the correlation energy and the exchange energy. In theory, the energy E_{XC} should also include the self–interaction energy and a correction to the kinetic energy.

$$E[\{\phi_i\}] = T[\{\phi_i\}] + J[\rho] + E_{\text{XC}}[\rho] + \int v(\mathbf{r})\rho(\mathbf{r})\mathrm{d}\mathbf{r} \tag{2.2.12}$$

The minimization of the energy is done by Lagrangian optimization with analogue conditions compared to Hartree–Fock. This leads to the Kohn–Sham operator and a new eigenequation, which is shown in Eq (2.2.13).

$$\hat{f}^{\text{KS}}(\mathbf{r}_i)\phi_k(\mathbf{r}_i) = \epsilon(\mathbf{r}_i)\phi_k(\mathbf{r}_i) \tag{2.2.13}$$

$$\hat{f}^{\text{KS}}(\mathbf{r}_i) = \hat{h}_i + 2\sum_{j=1}^{n} \hat{J}_j + \hat{v}_{\text{xc}} \tag{2.2.14}$$

In comparison to the Fock operator, $\hat{f}^{\text{KS}}(\mathbf{r}_i)$ includes an exchange–correlation term, instead of the exchange operator. With a basis set approach analogue to the HF model, an eigenvalue equation similar to the Roothaan–Hall equation is derived (s. Eq. (2.2.15)).

$$\mathbf{FC} = \mathbf{SC}\epsilon \tag{2.2.15}$$

2.2.3 GFN2–xTB Electronic Structure Method

A short introduction to GFN2–xTB electronic structure method is given in this part. The derivation is based on reference [15]. GFN2–xTB is a Density Functional Tight Binding (DFTB) method, that is optimized for **G**eomteries, **F**requencies and **N**oncovalent interaction energies. In DFTB methods, the electron density ρ_0 is a linear combination of atomic electron densities $\rho_{0,A}$.

$$\rho_0 = \sum_A \rho_{0,A} \tag{2.2.16}$$

The energy is expanded by a Taylor series around the electron density ρ_0. The bonding is then described as a perturbation to this density. In GFN2–xTB the expansion is done up to the third order.

$$E[\rho] = E^{(0)}[\rho_0] + E^{(1)}[\rho_0, (\delta\rho)] + E^{(2)}[\rho_0, (\delta\rho)^2] + E^{(3)}[\rho_0, (\delta\rho)^3] + \dots \tag{2.2.17}$$

In contrast to other DFTB methods, the derivation of the energy expansion for GFN2–xTB directly includes a non–local correlation contribution. Furthermore, the energy (s. Eq. (2.2.18)) is described with the kinetic energy $T[\rho(\mathbf{r})]$ and the external potential, due to the nuclei $V_n(\mathbf{r})$. A local density approximation (LDA) is used for the description of the exchange–correlation energy $\epsilon_{\mathrm{XC}}^{\mathrm{LDA}}$. As the LDA neglects the long range correlation effects, these are described via a non–local correlation contribution $\Phi_{\mathrm{C}}^{\mathrm{NL}}(\mathbf{r}, \mathbf{r}')$.

$$\begin{aligned} E[\rho] = \int \rho(\mathbf{r}) \Big[& T[\rho(\mathbf{r})] + V_n(\mathbf{r}) + \epsilon_{\mathrm{XC}}^{\mathrm{LDA}}[\rho(\mathbf{r})] \\ & + \frac{1}{2} \int \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} + \Phi_{\mathrm{C}}^{\mathrm{NL}}(\mathbf{r}, \mathbf{r}') \right) \rho(\mathbf{r}')\, \mathrm{d}\mathbf{r}' \Big] \mathrm{d}\mathbf{r} + E_{nn} \end{aligned} \tag{2.2.18}$$

The terms of Eq. (2.2.17) can be divided into different energy contributions. These contributions will be briefly discussed. The zeroth order term shown in Eq. (2.2.19) consists of the repulsion energy $E_{\mathrm{rep}}^{(0)}$ and dispersion energy $E_{\mathrm{disp}}^{(0)}$. These are typically described by Buckingham or Lennard–Jones potentials. In GFN2–xTB a classical parametrized repulsion energy is used, which depends on the distance between to atoms. The dispersion term arises, due to the non–local correlation functional. Furthermore, the zeroth order term energy of the valence electrons $E_{\mathrm{A,valence}}^{(0)}$ and core electron $E_{\mathrm{A,core}}^{(0)}$ is summed, though the energy of the core electrons is defined as zero.

$$E[\rho_0] \approx E_{\mathrm{rep}}^{(0)} + E_{\mathrm{disp}}^{(0)} + \sum_A \left(E_{\mathrm{A,valence}}^{(0)} + E_{\mathrm{A,core}}^{(0)} \right) \tag{2.2.19}$$

The first order perturbation energy $E^{(1)}[\rho_0,(\delta\rho)]$ is approximated by a first order dispersion energy $E^{(1)}_{\mathrm{disp}}$, and first order valence electron energy $E^{(1)}_{\mathrm{A,valence}}$. This perturbation allows the description of covalent bonding.

$$E^{(1)}[\rho_0,(\delta\rho)] \approx E^{(1)}_{\mathrm{disp}} + \sum_A E^{(1)}_{\mathrm{A,valence}} \tag{2.2.20}$$

The second order energy perturbation $E^{(2)}[\rho_0,(\delta\rho)^2]$ is described by the contribution of dispersion, electrostatics and exchange correlation energies. The exchange and correlation energy $E^{(2)}_{\mathrm{XC}}$ and electrostatic energy $E^{(2)}_{\mathrm{ES}}$ are described in GFN2–xTB with isotropic and anisotropic contributions. This is in contrast to other DFTB methods, where only a monopole approximation is used.

$$E^{(2)}[\rho_0,(\delta\rho)^2] \approx E^{(2)}_{\mathrm{ES}} + E^{(2)}_{\mathrm{XC}} + E^{(2)}_{\mathrm{disp}} \tag{2.2.21}$$

For the third order perturbation, only the electrostatic and exchange correlation energies are used and computed using the Mulliken approximation. All other contributions to the third order perturbation are neglected.

$$E^{(3)}[\rho_0,(\delta\rho)^3] \approx E^{(3)}_{\mathrm{ES}} + E^{(3)}_{\mathrm{XC}} \tag{2.2.22}$$

Merging all energy contributions shown in, Eq. (2.2.19) to Eq. (2.2.22), into Eq. (2.2.17) leads to Eq. (2.2.23). Though, some further adjustments are done. The isotropic contribution of the electrostatic and exchange correlation energies are described by a single term $E_{\mathrm{IES+IEC}}$. While the anisotropic electrostatic energy E_{AES} and anisotropic exchange correlation energy E_{AXC} are described separately. The valence electron energy contributions are described via the extended Hückel type energy E_{EHT}. Independent of the above discussed perturbation, an energy term G_{Fermi} describing Fermi–smearing is included, which leads to the complete energy equation shown in Eq. (2.2.23).

$$E_{\mathrm{GFN2-xTB}} = E_{\mathrm{rep}} + E_{\mathrm{EHT}} + E_{\mathrm{disp}} + E_{\mathrm{IES+IEC}} + E_{\mathrm{AES}} + E_{\mathrm{AXC}} + G_{\mathrm{Fermi}} \tag{2.2.23}$$

The wave function is approximated by a minimal valence basis set STO–mG. The energy is computed similar to KS–DFT and Hartree Fock. The Roothaan–Hall type Eq. (2.2.10) is solved in a selfconsistent manner.

2.3 Hessian Matrices

Hessian matrices are the target for the ML model. A brief introduction to the Hessian matrix, its projection and weighting is given, because it is necessary for the computation of the HVF. The HVF is used to compute quantities to measure the performance of the ML model.
The Hessian matrix is a $3M \times 3M$ matrix (M is the number of atoms) of second derivatives of the energy with respect to every atom coordinate pair in the system.[10]

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 E}{\partial x_A \partial x_A} & \cdot\cdot & \frac{\partial^2 E}{\partial x_A \partial y_M} & \frac{\partial^2 E}{\partial x_A \partial z_M} \\ \frac{\partial^2 E}{\partial x_A \partial y_A} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 E}{\partial z_M \partial x_A} & \cdot\cdot & & \frac{\partial^2 E}{\partial z_M \partial z_M} \end{pmatrix} \tag{2.3.1}$$

The Hessian $\mathbf{H}$ can also be described by M^2 3×3 block matrices, where each 3×3 matrix depends on the coordinates of an atom pair. Theses 3×3 block matrices are beneficial for the prediction, as they only depend on two atoms. Hence, only information about the two atoms need to be given to the ML model to predict the block matrix.

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{AA} & \mathbf{H}_{AB} & \cdot\cdot & \mathbf{H}_{AM} \\ \mathbf{H}_{BA} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \mathbf{H}_{MA} & \cdots & \cdot\cdot & \mathbf{H}_{MM} \end{pmatrix} \tag{2.3.2}$$

Due to the Schwartz Theorem, the $\mathbf{H}_{BA}$ and $\mathbf{H}_{AB}$ are related according to Eq. (2.3.3)

$$\mathbf{H}_{BA} = (\mathbf{H}_{AB})^{\intercal} \tag{2.3.3}$$

Therefore, the information of the Hessian can be stored as a triangular matrix (s. Eq. (2.3.4)).

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{AA} & \mathbf{H}_{AB} & \cdot\cdot & \mathbf{H}_{AM} \\ & \ddots & & \vdots \\ & 0 & \ddots & \vdots \\ & & & \mathbf{H}_{MM} \end{pmatrix} \tag{2.3.4}$$

The Hessian matrix needs to be projected for the computation harmonic frequencies. In order to compute the projection of the Hessian matrix, the rotational and translational parts of it need to be detected. A $6 \times 3M$ matrix is constructed, that describes the rotation and translation of each atom. For an atom A, the overlap is depicted in Eq. (2.3.5). The first three rows of the $\mathbf{O}_A$ matrix represents the translation and the latter three rows describe the rotation. For example, the rotation around the x–axis leads to the translation of the y–coordinate $\mathbf{r}_A^y$ by $\mathbf{r}_A^z$. While the z–coordinate $\mathbf{r}_A^z$ is moved by $-\mathbf{r}_A^y$.

$$\mathbf{O}_A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & \mathbf{r}_A^z & -\mathbf{r}_A^y \\ -\mathbf{r}_A^z & 0 & \mathbf{r}_A^x \\ \mathbf{r}_A^y & -\mathbf{r}_A^x & 0 \end{pmatrix} \tag{2.3.5}$$

For a system of M atoms, the full matrix $\mathbf{O}$ is shown in Eq. (2.3.6)

$$\mathbf{O} = \begin{pmatrix} \mathbf{O}_A & \mathbf{O}_B & \dots & \mathbf{O}_N \end{pmatrix} \tag{2.3.6}$$

After that, the rotation and translation matrix is normalized to $\mathbf{O}_\mathrm{n}$ by the 1–norm. For the detection of the translation and rotation contributions, the overlap matrix $\mathbf{M}$ between the normalized rotation and translation matrix and the Hessian matrix is computed (s. Eq. (2.3.7)). The multiplication of the $6 \times 3M$ and $3M \times 3M$ matrix leads to a $6 \times 3M$ matrix.

$$\mathbf{M} = \mathbf{O}_\mathrm{n} \times \mathbf{H} \tag{2.3.7}$$

The columns of $\mathbf{M}$ are summed up, which leads to a $3M$ vector. The maximum values of this vector corresponds to the highest overlap between a row in the Hessian matrix and the translation and rotation matrix. The six highest values are searched, and the indices of the corresponding row are stored in a vector $\mathbf{v}$. For a linear molecule only the five highest values are searched. The differentiation between linear and non–linear molecules is done by the norms of the coordinates $\|\mathbf{r}^x\|$,$\|\mathbf{r}^y\|$ and $\|\mathbf{r}^z\|$. If the sum of the norm of two coordinates is below the threshold of $\|\mathbf{r}^{x,y,z}\| < 1 \cdot 10^{-6}$ Å a linear molecule is assumed.

For the projection itself (s. Eq. (2.3.8)), the eigenvectors $\mathbf{Q}$ of the Hessian matrix and its eigenvalues λ are required.

$$\mathbf{H}^{\text{proj}} = \mathbf{H} - \sum_{i \in \mathbf{v}} \lambda_i \cdot \mathbf{Q}^{\mathsf{T}}{}_i \otimes (\mathbf{Q}^{\mathsf{T}}{}_i)^{\mathsf{T}} \tag{2.3.8}$$

At last, the Hessian needs to be mass weighted.

$$\mathbf{H}_{AB}^{\text{proj,w}} = \frac{1}{\sqrt{m_A m_B}} \mathbf{H}_{AB}^{\text{proj}} \tag{2.3.9}$$

Eq. (2.3.9) is applied to every Hessian matrix block $\mathbf{H}_{AB}$ in the full Hessian matrix $\mathbf{H}$, which leads to the projected mass weighted Hessian matrix $\mathbf{H}^{\text{proj,w}}$.

2.4 Machine Learning

This section deals with basics of the ML algorithms used in this work and is based on reference [16]. These ML alogrithms are the Extra Tree Regression (ETR) and Random Forest Regression (RFR). Both algorithms are ensemble methods based on decision trees as individual estimators. In this section, decision tree based algorithms will be discussed, and the differences between the ETR and RFR will be emphasized.

Often these decision tree based algorithms are based on the Classification and Regression Tree (CART) algorithm. In the example of a response function Y and its variables $X1$ and $X2$, Y is the target and $X1$ and $X2$ are the features. With the CART algorithm, a tree based is constructed. This tree based model splits the space of this function into several recursive binary partitions. Recursive binary means in this case, that the splits are orthogonal to the axis of the features. This split is shown on the left in Fig. 2.2.

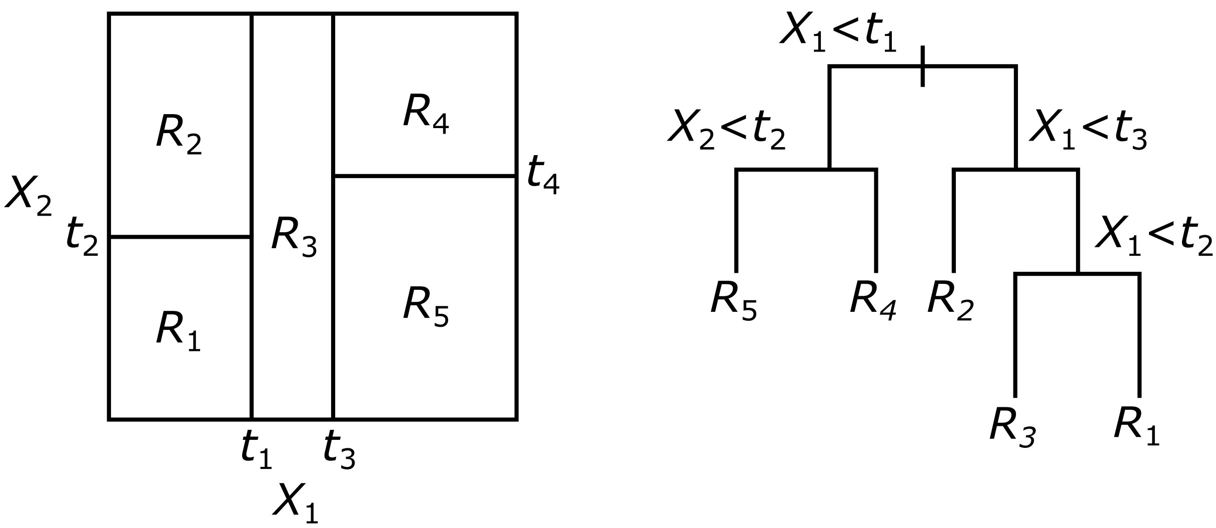


Figure 2.2: The recursively binary grown split is shown on the left. On the right, the resulting Decision Tree is depicted.

In each partition, Y is modeled with a function or constant R_i. This can also be depicted in a more familiar approach, the decision tree. This is shown in Fig. 2.2. Both depictions lead to the same result.
The splitting points t_1 to t_4 are chosen differently for the RFR and ETR. In RFR, the splitting point is determined by choosing the best feature and of this feature the best split value. Hence, it is a two dimensional search. In the case of the ETR, the feature is chosen randomly. After that, the best split value for this feature is chosen. The relation between the features and the partitioned response Y can be depicted in a decision tree, which is shown in Fig. 2.2 on the right. This can also be described mathematically via Eq. (2.4.1)

$$\hat{f}(X) = \sum_{m=1}^{5} c_m I\{(X_1, X_2) \in R_m\} \tag{2.4.1}$$

Here, $\hat{f}(X)$ is the function that describes the response Y. Each region m has a corresponding constant c_m. The function I can either be $I = 1$, if the features X_1 and X_2 are in the rectangle R_m, or it is $I = 0$, if it is not the case. Originally, the ETR and RFR have another difference, which is that for ETR bootstrapping was not used.[17] Bootstrapping uses the training set and makes several new training sets from this. The new training sets are randomly filled with the datasets of the original training set.[16] This differentiation is not necessarily done anymore. In software like `sklearn`[18] it is optional to use bootstrapping.

3 Methods

3.1 Transformation of Coordinates

The Hessian matrix and the features that originate from vector or matrix quantities depend on the orientation of the coordinate system (CS). For the ML model a consistent orientation for every atom pair is necessary. Furthermore, the Hessian elements shall be learned atom pair wise. Thus, the orientation of the molecule is brought into atom pair focused CS (APF CS), where the origin of the CS is in the center of the considered atom pair, while both atoms are aligned on the z–axis. In this section, the transformation of the initial coordinate system to the APF CS will be discussed.
The CS of the molecule is deliberate, therefore a transformation to a well–defined reference coordinate system is necessary. As a reference coordinate system the main inertia coordinate system (MI CS) is chosen. In a second step the transformation to the APF CS is performed.
At first a translation of the origin of the initial CS to the center of mass is done by Eq. (3.1.1).

$$\mathbf{r}_A^{\mathrm{s}} = \mathbf{r}_A^{\mathrm{init}} - \mathbf{s}^{\mathrm{init}} \tag{3.1.1}$$

Where $\mathbf{r}_A^{\mathrm{s}}$ is the coordinate vector of an atom A in the mass centered CS, $\mathbf{r}_A^{\mathrm{init}}$ the coordinate vector in the initial CS and $\mathbf{s}_A^{\mathrm{init}}$ is the center of mass in the initial CS, and is defined in Eq. (3.1.2).

$$\mathbf{s}^{\mathrm{init}} = \frac{1}{m_{\mathrm{tot}}} \sum_{A}^{M} m_A \cdot \mathbf{r}_A^{\mathrm{init}} \tag{3.1.2}$$

Here m_A is the mass of atom A, m_{tot} is the mass of the molecule and M the number of atoms in the molecule.

Once the origin of the new CS is determined, the moment of inertia is computed by Eq. (3.1.3).

$$\mathbf{I}^{\mathrm{s}} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix}; I_{ab} = \begin{cases} \sum_A^M m_A(b_A^2 + c_A^2) & \text{if } a = b \\ -\sum_A^M m_A \cdot a_A \cdot b_A & \text{if } a \neq b \end{cases} \quad a, b \in x, y, z \tag{3.1.3}$$

For the computation of the elements I_{ab} of the inertia tensor $\mathbf{I}^{\mathrm{s}}$ the diagonal elements and non–diagonal elements are divided in two cases. Here, a and b are variables for the spatial directions of the coordinates x, y and z.

For the rotation of the mass centered origin CS into the MI CS, the matrix $\mathbf{R}_{\mathrm{mi}}^{\mathrm{init}}$ of eigenvectors of the inertia tensor $\mathbf{I}^{\mathrm{s}}$ is needed. For that, the eigenvectors $\mathbf{R}_{\mathrm{mi}}^{\mathrm{init}}$ are computed by solving the eigenvalue equation. The eigenvalues are then used to get the eigenvectors. To assure, that the matrix $\mathbf{R}_{\mathrm{mi}}^{\mathrm{init}}$ is a right–handed matrix the determinant is calculated. If the determinant is negative, the sign of the third eigenvector is changed. Furthermore, it needs to be assured, that the eigenvector matrix is consistently rotated in the same direction, independent of the outgoing position. The chosen constraint is, that the highest value of the eigenvectors are positive, if this is not the case the sign of the eigenvector and the next eigenvector in $\mathbf{R}_{\mathrm{mi}}^{\mathrm{init}}$ is changed.
The rotation of the coordinates is done by Eq. (3.1.4).

$$\mathbf{r}_A^{\mathrm{mi}} = \mathbf{R}_{\mathrm{mi}}^{\mathrm{init}} \mathbf{r}_A^{\mathrm{s}} \tag{3.1.4}$$

Here, $\mathbf{r}_A^{\mathrm{mi}}$ is the vector of the coordinates of an atom A in the MI CS. The rotation is done for every atom in the molecule. After that operation, the molecule is in the MI CS.

The Hessian matrix also depends on the CS and therefore needs to be rotated, too. A translation is not needed. Eq. (3.1.6) describes the rotation of the Hessian matrix in the initial CS to the MI CS. For this a $3M \times 3M$ rotation matrix $\mathbf{P}_{\text{mi}}^{\text{init}}$ needs to be constructed, where the diagonal 3 x 3 blocks are $\mathbf{R}_{\text{mi}}^{\text{init}}$. This matrix is shown in Eq. (3.1.5).

$$\mathbf{P}_{\text{mi}}^{\text{init}} = \begin{pmatrix} \mathbf{R}_{\text{mi}}^{\text{init}} & \mathbf{0} & & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{\text{mi}}^{\text{init}} & & \mathbf{0} \\ \mathbf{0} & & \ddots & \\ & & \ddots & \mathbf{0} \\ \mathbf{0} & & \mathbf{0} & \mathbf{R}_{\text{mi}}^{\text{init}} \end{pmatrix} \tag{3.1.5}$$

$$\mathbf{H}^{\text{mi}} = \mathbf{P}_{\text{mi}}^{\text{init}} \times \mathbf{H}^{\text{init}} \times \left(\mathbf{P}_{\text{mi}}^{\text{init}}\right)^{\text{T}} \tag{3.1.6}$$

The Hessian matrix in the initial CS is $\mathbf{H}^{\text{init}}$, while the Hessian matrix in the MI CS is $\mathbf{H}^{\text{mi}}$.
There are vector and matrix quantities that are used as features for the ML models. These features and the Hessian matrix depend on their orientation in the CS. Moreover, the Hessian matrix elements are predicted element–wise. Hence, it is necessary to use a consistent atom pair specific CS. For that the MI CS is transformed into the APF CS. The transformation starts with the translation of the origin of the new CS to the center of an atom pair. The translation is described with Eq. (3.1.7).

$$\mathbf{r}_C^{\text{mi,c}} = \mathbf{r}_C^{\text{mi}} - \left(\frac{1}{2}\mathbf{r}_A^{\text{mi}} + \frac{1}{2}\mathbf{r}_B^{\text{mi}}\right) \tag{3.1.7}$$

After this operation, the origin of the CS is in the center of the atom pair. To fix the orientation, the atom pair is aligned along the z–axis. The Euler angles are used to determine the rotation matrix. The rotation of the CS depends on the considered atom pair. It is divided into whether $A = B$ or $A \neq B$. At first the construction of the rotation matrix for the case $A \neq B$ will be discussed.
The z–axis of the APF CS is aligned to $\mathbf{r}_A^{\text{mi,c}}$. The x–axis is defined in Eq. (3.1.8), where $\mathbf{z}$ is a unit vector in z–direction of the APF CS, and $\boldsymbol{\mu}_A$ is the dipole moment of the atom A.

$$\mathbf{x} = \mathbf{z} \times (\boldsymbol{\mu}_{\text{A}} + \boldsymbol{\mu}_{\text{B}}) \tag{3.1.8}$$

The intersection $\mathbf{L}$ of the xy–plane of the APF CS and xy–plane of the centered MI CS is described with Eq. (3.1.9).

$$\mathbf{L} = \mathbf{x} \times \mathbf{z}^{\mathrm{mi,c}} \tag{3.1.9}$$

The intersection $\mathbf{L}$, the axes $\mathbf{x}, \mathbf{z}, \mathbf{x}^{\mathrm{mi,c}}$ and $\mathbf{z}^{\mathrm{mi,c}}$ are then used for the computation of the Euler angles α^0, β and γ^0 via Eq. (3.1.10).

$$\alpha^0 = \arccos\left(\frac{\mathbf{L} \cdot \mathbf{x}^{\mathrm{mi,c}}}{|\mathbf{L}| \cdot |\mathbf{x}^{\mathrm{mi,c}}|}\right) \tag{3.1.10a}$$

$$\beta = \arccos\left(\frac{\mathbf{x} \cdot \mathbf{z}^{\mathrm{mi,c}}}{|\mathbf{x}| \cdot |\mathbf{z}^{\mathrm{mi,c}}|}\right) \tag{3.1.10b}$$

$$\gamma^0 = \arccos\left(\frac{\mathbf{L} \cdot \mathbf{x}}{|\mathbf{L}| \cdot |\mathbf{x}|}\right) \tag{3.1.10c}$$

For the right rotation, it has to be considered, whether the vectors $\mathbf{x}^{\mathrm{mi,c}}, \mathbf{z}^{\mathrm{mi,c}}$ and $\mathbf{L}$ span a right–handed coordinate system or left–handed. This is done by constructing a matrix of these vectors and computing the determinant. The determinant can be used, as it is equal to the triple product for a 3×3 matrix.

$$\mathbf{A} = (\mathbf{x}, \mathbf{z}, \mathbf{L}) \tag{3.1.11}$$

$$\mathbf{G} = (\mathbf{L}, \mathbf{x}^{\mathrm{mi,c}}, \mathbf{z}^{\mathrm{mi,c}}) \tag{3.1.12}$$

$$\alpha = \begin{cases} \alpha^0 & \text{if } \det(\mathbf{A}) = 1 \\ 2\pi - \alpha^0 & \text{if } \det(\mathbf{A}) = -1 \end{cases} \tag{3.1.13}$$

$$\gamma = \begin{cases} \gamma^0 & \text{if } \det(\mathbf{G}) = -1 \\ 2\pi - \gamma^0 & \text{if } \det(\mathbf{G}) = 1 \end{cases} \tag{3.1.14}$$

Via these constraints, the Euler angles α and γ are chosen. The Euler angles are then used to build up the rotation matrices via Eq. (2.1.1) and (2.1.2).

The complete rotation matrix is then constructed by the cross product of the single rotation matrices and is shown in Eq. (3.1.15).

$$\mathbf{R}^0 = \mathbf{R}_z(\gamma) \times \mathbf{R}_x(\beta) \times \mathbf{R}_z(\alpha) \tag{3.1.15}$$

If the x–values of the rotated dipole moment $\boldsymbol{\mu}$ are negative, the matrix is rotated by 180° around the z–axis. This is depicted in Eq. (3.1.16).

$$\mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}} = \begin{cases} \mathbf{R}^0 & \text{if } \boldsymbol{\mu}^x > 0 \\ \mathbf{R}^0 \times \mathbf{R}_z(\pi) & \text{if } \boldsymbol{\mu}^x < 0 \end{cases} \tag{3.1.16}$$

Lastly, the rotation of the translated coordinate system is done via Eq. (3.1.17).

$$\mathbf{r}_A^{\mathrm{apf}} = \mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}} \times \mathbf{r}_A^{\mathrm{mi,c}} \tag{3.1.17}$$

The coordinates $\mathbf{r}_A^{\mathrm{apf}}$ are the final coordinates in the APF CS for the case of a heteronuclear atom pair.

For the case of a homonuclear atom pair, the procedure is the same up to the translation of the coordinates into the centered MI CS. The z–axes of APF CS and centered MI CS are aligned. Therefore, the Euler angle β is zero. Furthermore, the xy–planes are the same, and there is no intersection of the xy–planes. Thus, only one angle α or γ is necessary for the rotation. Therefore, γ is the angle between the x–axis of the centered MI CS and the APF CS. To provide consistency in the rotation of the coordinates, the x–axis is chosen in dependency of the dipole moment $\boldsymbol{\mu}$ of the considered atom. In Eq. (3.1.18) is shown how the x–axis is computed.

$$\mathbf{x} = \mathbf{z} \times \boldsymbol{\mu}_A \tag{3.1.18}$$

This results in the computation of the angle γ^0 by Eq. (3.1.19).

$$\gamma^0 = \arccos\left(\frac{\mathbf{x}^{\mathrm{mi,c}} \cdot \mathbf{x}}{|\mathbf{x}^{\mathrm{mi,c}}| \cdot |\mathbf{x}|}\right) \tag{3.1.19}$$

Between two vectors are two angles, which sum up to 2π. The rotation of the CS depends on the arrangement to another. The angle is chosen by the condition shown in Eq. (3.1.20).

$$\gamma = \begin{cases} \gamma^0 & \text{if } \mathbf{x}_y < 0 \\ 2\pi - \gamma^0 & \text{if } \mathbf{x}_y \geq 0 \end{cases} \tag{3.1.20}$$

The further procedure of the transformation is done in the same way as for the heteronuclear atomic pair. The rotation matrix is computed via Eq. (3.1.15). This is followed by an adjustment of the rotation matrix, shown in Eq. (3.1.16). The final rotation matrix $\mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}}$ is then used on the coordinates $\mathbf{r}_A^{\mathrm{mi,c}}$ via Eq. (3.1.17).

In the APF CS the computation of the Hessian was done. For each APF CS one 3×3 Hessian matrix block is of interest. To rebuild the Hessian matrix in the MI CS, each Hessian matrix block needs to undergo a rotation back to the MI CS. For this the all rotation matrices $\mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}}$ need to be stored. The rotation back into the MI CS is done by Eq. (3.1.21).

$$\mathbf{H}_{\mathrm{AB}}^{\mathrm{mi}} = \left(\mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}}\right)^{\mathsf{T}} \times \mathbf{H}_{\mathrm{AB}}^{\mathrm{apf}} \times \mathbf{R}_{\mathrm{mi}}^{\mathrm{apf}} \tag{3.1.21}$$

In the MI CS, the Hessian matrix $\mathbf{H}^{\mathrm{mi}}$ is filled with the blocks $\mathbf{H}_{\mathrm{AB}}^{\mathrm{mi}}$. Before some quantities for measuring can be computed, the $\mathbf{H}^{\mathrm{mi}}$ is projected and mass–weighted. Thus, the $\mathbf{H}^{\mathrm{proj,w|mi}}$ is build. The procedure of the projection and mass–weighting is described in the chapter Theory.

3.2 Machine Learning Model Construction

Supervised ML models have to be trained with a set of features and targets. In this work, the features are mostly derived from atomic quantities based on geometry, electron density and energy. These feature selection criteria are based on Steinbach's work.[19] Some adaptations are done. Namely, the vector of the dipole moment instead of its norm is used. Furthermore, instead of the norm of the quadrupole a vector derived from the quadrupole matrix is used. The use of the vectors and matrix is one reason for the transformation of the CS. The atomic quantities and Hessian matrix are generated with the GFN2–xTB method. The quantities computed with GFN2–xTB are listed in Tab. 3.1 with a short description. There is an extended version for some quantities. If this is the case, the column Extension is ticked. The mathematical descriptions of all quantities and its extension from GFN2–xTB are shown in the Appendix in Eq. (7.0.1)

Table 3.1: List of quantities used as features for the ML models. The quantities are divided into geometry, density and energy based quantities.

Feature	Description	Extension
Geometry		
R_{AB}	Distance between two Atoms	
CN_A	Coordination Number	✓
Density		
μ_A	Dipole Moment Vector	✓
$\mu_{A,\mathrm{p}}$	Dipole Moment Vector on A, due to distribution on p and μ on other atoms	
θ_A	Vector computed from Quadrupole Moment Matrix	✓
Energy		
$E_{A,\mathrm{gap}}$	HOMO – LUMO gap	✓
$\mu_{A,\mathrm{pot}}$	Chemical Potential	✓
$\epsilon_{A,\mathrm{HOAO}}$	Atomic HOMO	✓
$\epsilon_{A,\mathrm{LUAO}}$	Atomic LUMO	✓
$E_{A,\mathrm{rep}}$	Repulsion	
$E_{A,\mathrm{EHT}}$	extended Hückel theory	
$E^{(2)}_{A,\mathrm{disp}}$	Dispersion 2$^{\mathrm{nd}}$ order	
$E^{(3)}_{A,\mathrm{disp}}$	Dispersion 3$^{\mathrm{rd}}$ order	
$E_{A,\mathrm{IES+IXC}}$	Isotropic ES and XC	
$E_{A,\mathrm{AES}}$	Anisotropic ES	
$E_{A,\mathrm{AXC}}$	Anisotropic XC	
$E_{A,\mathrm{tot}}$	Total Energy	

Either the atomic quantities are used directly as the atomic features, or are further processed. Beside atomic quantities, further features are implemented for the differentiation between the elements of a block $\mathbf{H}_{AB}$. The features are computed for every Hessian matrix block $\mathbf{H}_{AB}$ in its specific CS. Only the atom based quantities of A and B are used for the computation of the features.

Two different models are used for $A = B$ and $A \neq B$, due to two reasons. At first the chosen orientation in the CS between those cases differ. Secondly the feature vectors have a different size. For the case $A \neq B$, the features are computed from the GFN2–xTB quantities using Eq. (3.2.1). Where x is a placeholder for any of the quantities shown in Tab. 3.1, but R_{AB}. This case is called **heteronuclear**, as the Hessian and features depend on two atoms.

$$x_{\text{Arith}} = \frac{x_A + x_B}{2} \tag{3.2.1a}$$

$$x_{\text{Prod}} = x_A \cdot x_B \tag{3.2.1b}$$

$$x_{\text{AbsDiff}} = |x_A - x_B| \tag{3.2.1c}$$

In the case of $A = B$ the quantities are used directly, and it is called **homonuclear**. Here, the Hessian matrix elements and features only correspond to one atom and R_{AB} is not used.

Connecting the Hessian elements to the features leads to the following problem. The features for all elements in a Hessian matrix block $\mathbf{H}_{AB}$ are the same. Therefore, the ML model would predict the same value for every matrix element. Hence, a differentiation is needed for the elements.

Two approaches are used for the differentiation. One approach (indexed/Idx) is the use of three further features x_i, y_i, z_i. These are indices, which differentiate between each matrix element. In Tab. 7.1 in the Appendix the values for those indices as a function of the matrix elements is shown.

Another approach (permuted/Perm) is the permutation of the coordinate system and with it the vector/matrix quantities from GFN2–xTB, in dependence of the Hessian elements that shall be learned. The coordinate system is always adjusted, so that the to be learned Hessian element is on the spot of the element $\frac{\partial^2 E}{\partial y_A \partial x_A}$. For example, if the element $\frac{\partial^2 E}{\partial x_A \partial z_A}$ is to be learned. The coordinate system is permuted from $\{x, y, z\}$ to $\{z, x, -y\}$. The Hessian matrix elements and features are permuted. The permutation of the vector quantities is shown in Eq.(3.2.2), where q_a are the vector elements. The vector quantities are $\boldsymbol{\mu}_A$ and $\boldsymbol{\mu}_{A,\mathrm{p}}$.

$$\begin{pmatrix} q_x \\ q_y \\ q_z \end{pmatrix} \longrightarrow \begin{pmatrix} q_z \\ q_x \\ -q_y \end{pmatrix} \tag{3.2.2}$$

The matrix quantities are permuted as shown in Eq. (3.2.3). Here, Q_{AA} are the matrix elements. The matrix quantities are the Hessian matrix blocks $\boldsymbol{H}_{AB}$ and the quadrupole moment $\boldsymbol{\Theta}_A$. From the permuted $\boldsymbol{\Theta}_A$ the features θ_A are computed as shown in Eq. (7.0.1).

$$\begin{pmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{yx} & Q_{yy} & Q_{yz} \\ Q_{zx} & Q_{zy} & Q_{zz} \end{pmatrix} \longrightarrow \begin{pmatrix} Q_{zz} & Q_{zx} & -Q_{zy} \\ Q_{xz} & Q_{xx} & -Q_{xy} \\ -Q_{yz} & -Q_{yx} & Q_{yy} \end{pmatrix} \tag{3.2.3}$$

This has in principle the advantage, that the most important vector quantity is always the same. The permutation is done in a manner, that the new coordinate system is right–handed. All permutations are described in the appendix in Tab. 7.2.

For the homonuclear this concludes in 37 or 35 features for each Hessian matrix element for the indexed or permuted case, respectively. And for the heteronuclear model there are 106 or 104 features for each Hessian element for the indexed or permuted case, respectively.

3.3 Measures for Quantifying the Performance

For first insights on the correlation of the target on the features, the Pearson Correlation Coefficient R^2 (s. Eq. (3.3.1)) can be computed. It describes the linear correlation between the target and the feature. Here, x is a feature, y the target and σ_x, σ_y the standard deviations and $\bar{x}, \bar{y}$ the mean of the feature or target, respectively. The target y corresponds to the Hessian elements. The values for R^2 are in the range of $0 \leq R^2 \leq 1$.

$$R^2 = \left(\frac{\sum_i (x_i - \bar{x}) \cdot \sum_i (y_i - \bar{y})}{\sigma_x \cdot \sigma_y} \right)^2 \tag{3.3.1}$$

For the quantification of the learning process of the ML models the mean squared error MSE (s. Eq. (3.3.2)) was chosen. The ML models are trained with GFN2–xTB based target and features. Therefore, the reference is also the GFN2–xTB computed Hessian. The number of data points is n.

$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_{i,\mathrm{ML}} - y_{i,\mathrm{xTB}})^2 \tag{3.3.2}$$

For the comparison of the different models (e.g. permuted and indexed) a normal distribution is assumed (s. Eq. (3.3.3)). For this the difference of a quantity q between the GFN2–xTB computation and ML model prediction is calculated. Furthermore, the standard deviation σ and mean $\overline{\Delta q}$ from the difference Δq is computed.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{\Delta q - \overline{\Delta q}}{\sigma} \right)^2 \right) \tag{3.3.3}$$

The investigated quantities q are the Hessian matrix elements and the wave number $\tilde{\nu}$ (s. Eq. (3.3.4a)). In addition to the Hessian matrix elements and $\tilde{\nu}$, the force constant k (Eq. (3.3.4b)) was examined. Though, k was used directly for comparisons. Here, λ are the eigenvalues of the mass weighted projected Hessian matrix $\mathbf{H}^{\mathrm{proj,w|mi}}$ and m_i the mass of each atom. For the transformation into SI–units, the Hartree Energy E_h, reduced Planck constant $\hbar$, speed of light c and the atomic mass unit u are required.

$$\tilde{\nu} = \sqrt{|\lambda|} \cdot \frac{E_\mathrm{h}}{\hbar \cdot c} \tag{3.3.4a}$$

$$k = \frac{E_\mathrm{h}^2}{\hbar^2 \cdot u} \cdot \sum_{i}^{M} \frac{1}{m_i} \lambda \tag{3.3.4b}$$

3.4 Computational Methods

All electronic structure computations were performed with GFN2–xTB[15], for this `xtb`[20] was used. The keywords "gfn2", "hess" and "ml_feature" were passed for the computation of the features and target. The computations were done for every structure in the APF CS. For transformation of the coordinates and the ML models python was used. Especially, the packages `sklearn`[18], `numpy`[21] and `pandas`[22,23] were used for this project. For the depiction of every graph the package `matplotlib`[24] was used. The versions of the used packages are shown in Tab. 3.2.

Table 3.2: Python modules used for the computation, data storing, constructing the ML model and plotting graphs.

Package	Version
pandas	1.4.2
numpy	1.22.3
sklearn	1.0.2
matplotlib	3.5.2

For ML models, the algorithms `RandomForestRegressor` and `ExtraTreeRegressor`, which origin from `sklearn`, were used. The default parameters were used, except the number of estimators was set `n_estimators` = 300 for the homonuclear model, and `n_estimators` = 100 for the heteronuclear model. No bootstrapping was used in this work. To achieve the split of the target and features in dependence each structure, `GroupShuffleSplit` from `sklearn` was used. For the algorithms and the split ten random seeds `random_state` $\in \{0, 22, 42, 100, 63, 54, 96, 35, 408, 74\}$ were used.

4 Results and Discussion

4.1 Machine Learning Models and Data Set

For every prediction of the Hessian matrix, it is necessary to combine the homonuclear and heteronuclear model. Three aspects of those models are further investigated, namely the algorithms, the labeling and a data set differentiation.

At first, as described in chapter 2, the RFR and ETR are used as the different algorithms for fitting the ML model. Secondly, there are two possibilities for the labeling. Either the indexed (Idx) case is used or the permuted (Perm) case. The third adjustment is the differentiation of data sets with a case specific (Case) and a general (Gen) model. For the case specific models, only one molecular system with different structures is used for the training of a model. While in the general model all molecular systems are used to train one single model.

For the further discussion, abbreviations are used for the description of the different variations of the model. As an example, the abbreviation for the model with the ETR as the algorithm, the indexed labeling and general case is ETR–Gen–Idx.

The data set for benchmarking the ML model consists of eight different molecules: H_2, N_2, O_2, F_2, HF, CO, H_2O, and NH_3. For each molecule, structures with different bond lengths are generated. Only the distance of one bond is changed for each molecule. This leads to 103 different structures. These structures are split into a training set of 75 % and a test set of 25 %. The split is done structure wise to achieve the possibility to compare other quantities than the Hessian elements. For each structure, the Hessian matrix and features are calculated with the GFN2–xTB method.

Moreover, the performance of the model to a complete unknown system is of interest. Therefore, a H_3–system is used. The full data set introduced above is used as a training set for the model, while the H_3–system is used as the test set.

4.2 Feature–Target Correlation

To start of, first investigations are done on the linear correlation between the features and target. Thus, the Pearson Correlation Coefficient R^2 is computed for each feature. The model Gen–Idx is used. No algorithm is used, as only the correlation of the computed target and features is investigated. The range of the coefficient is $0 \leq R^2 \leq 1$. The ten features with the highest scores are shown in Fig. 4.1 for the homonuclear and heteronuclear model.

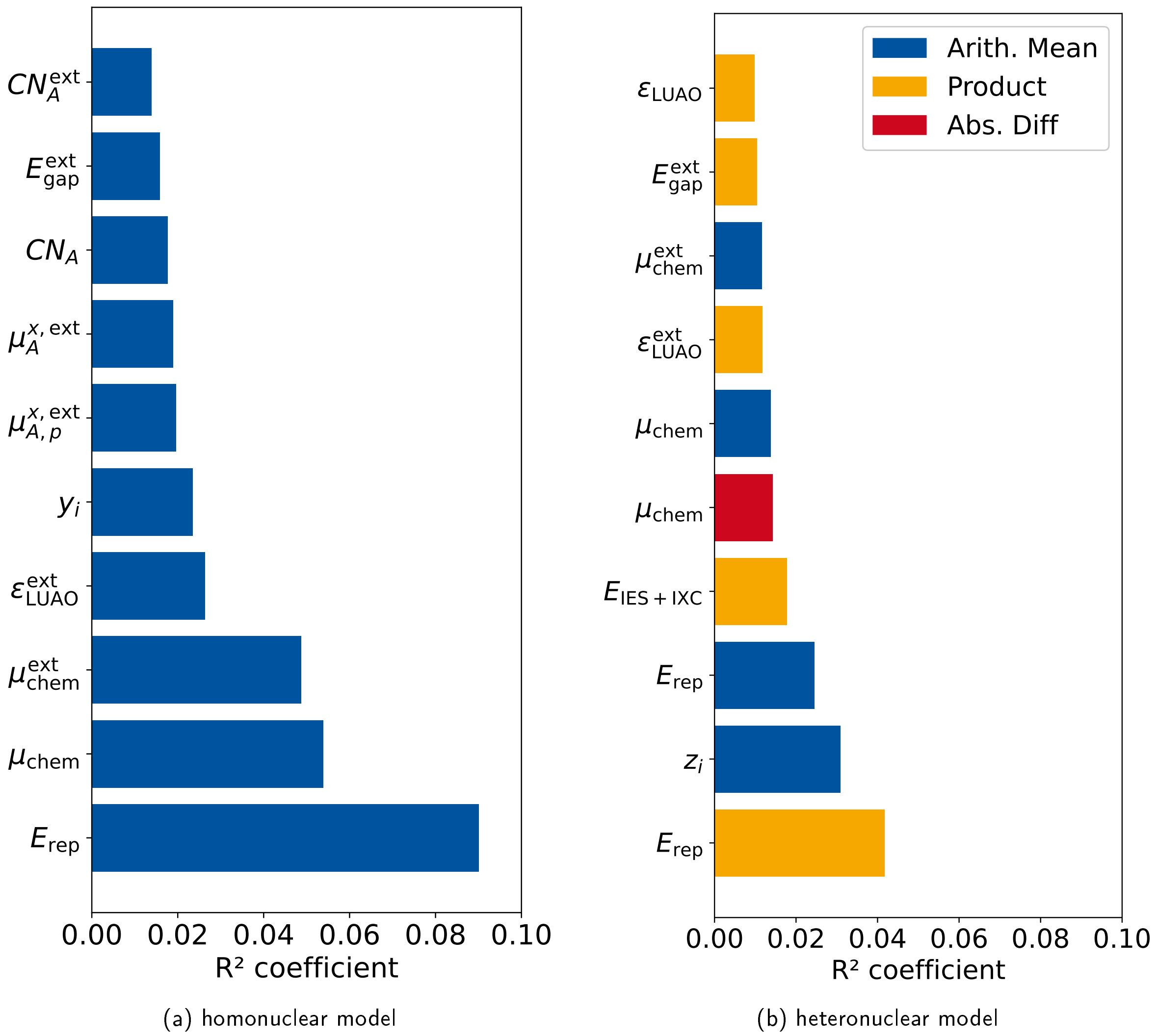
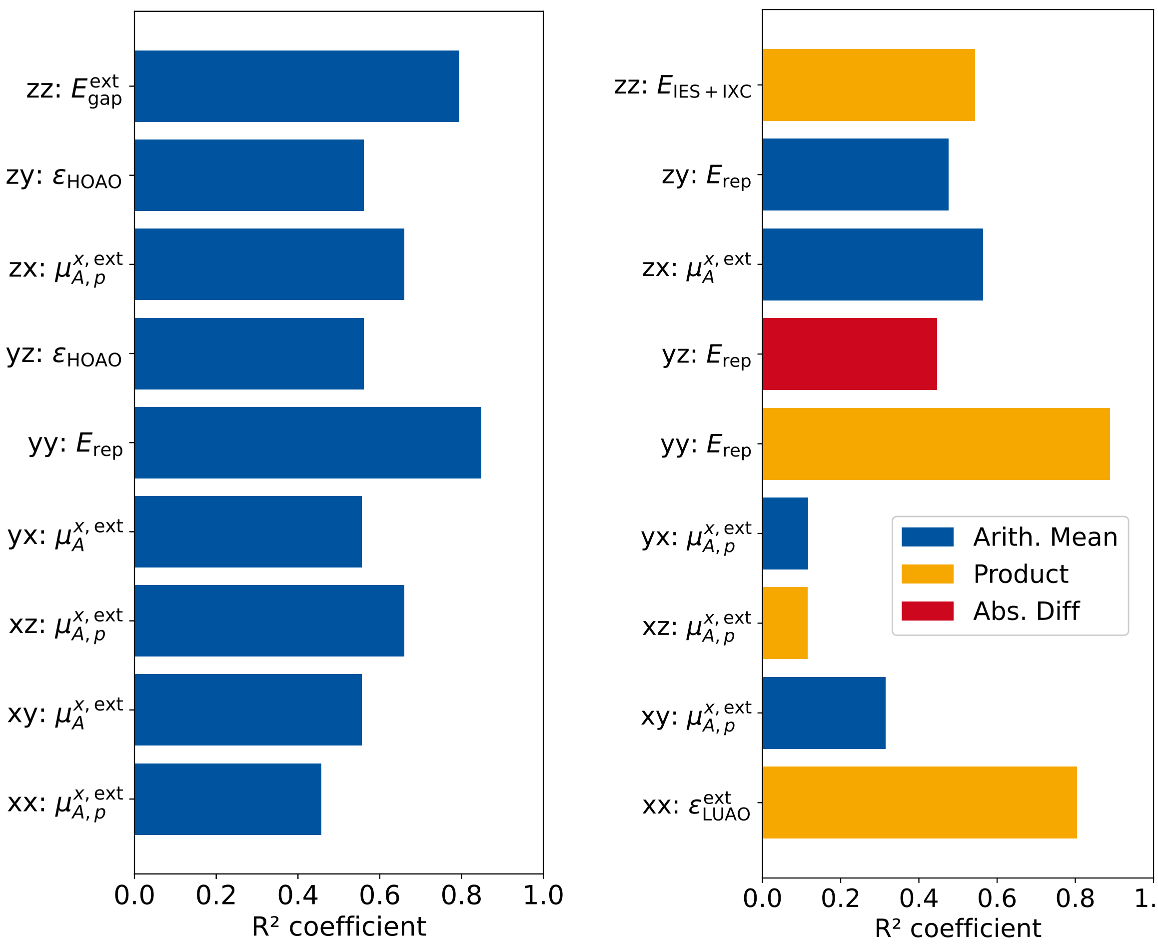


Figure 4.1: Pearson Correlation Coefficient for the features used for the homonuclear and heteronuclear model. The ten features with the highest coefficients are shown for each model.

The highest correlation in the homonuclear model is found for E_{rep}. For the heteronuclear model, the product of the E_{rep} has the highest correlation and third highest is its arithmetic mean. Hence, the repulsion seems to be an important feature for the prediction of the Hessian elements. Another high correlation feature is the chemical potential μ_{chem}. While, the potential itself has the second–highest correlation, its extension has the third highest for the homonuclear model. Furthermore, the three variations of μ_{chem} appear also in the heteronuclear model (5., 6., 8.).

At last, variations of E_{gap} and ϵ_{LUAO} are represented among the highest correlations for both models. For both models one index (y_i or z_i) is also among the ten highest correlations. While only the homonuclear model shows a high correlation with the coordination number CN_A and the atomic dipole moments μ, the heteronuclear shows a correlation with the atomic isotropic exchange and correlation energy $E_{\mathrm{IES+IXC}}$. However, the correlations are in general rather small. One possible reason is, that the Hessian matrix is sparse, therefore many matrix elements are $\frac{\partial^2 E}{\partial a_A \partial b_B} \approx 0$ and do not correlate with the given features. This can reduce R^2 significantly. Furthermore, the correlation is computed for all Hessian elements. If the feature correlations of the elements differ from another, the overall correlation is also decreased.

To investigate the last aspect, the target vector is split Hessian element–wise. For each Hessian element $\frac{\partial^2 E}{\partial a_A \partial b_B}$ the R^2 of the features is computed. For obvious reasons, the indices x_i, y_i and z_i are not included for this correlation. Fig 4.2 shows the highest R^2 for each Hessian element for the homo– and heteronuclear model. The elements are abbreviated with the coordinates of the derivative.



(a) homonuclear model

(b) heteronuclear model

Figure 4.2: Highest Pearson Correlation Coefficient specifically computed for each Hessian element for the homo– and heteronuclear model. The abbreviation ab corresponds to the elements $\frac{\partial^2 E}{\partial a_A \partial b_B}$.

Comparing the highest R^2 in Fig 4.1 with R^2 in Fig 4.2, there is a significant increase of R^2 for the homonuclear and heteronuclear model. For the heteronuclear model, there are two exceptions, namely for yx and xz. Though the element–wise R^2 is more than a magnitude larger than for the general correlation, the R^2 for yx and xz are only about the factor two to three larger than the R^2 of the general correlation. Interesting to point out, is that most of the features which show element–wise the highest correlation are also presented in the ten highest R^2 for the general correlation. Exceptions are for the homonuclear model ϵ_{HAOA}, and for the heteronuclear model ${\mu_A^{\mathrm{x,ext}}}_{\mathrm{Arith}}$, ${\mu_{A,p}^{\mathrm{x,ext}}}_{\mathrm{Prod}}$, ${E_{\mathrm{rep}}}_{\mathrm{Abs.Diff}}$. Regarding only the element–wise R^2 for the homonuclear model, the element pairs (xy,yx), (zx,xz) and (zy,yz) have the same correlation. This is due to the fact, that the Hessian matrix block $\mathbf{H}_{AA}$ is symmetric. Moreover, the highest correlating features do differ from another for each element, except for the symmetric element pairs. This is especially important for the ML, as this suggests, that the elements can be differentiated with the used features. This analysis is also done for the model Gen–Perm and shown in Fig 7.1 and Fig 7.2.

4.3 The H_2 System

The ETR–Case–Perm model is used as a first case for investigations with the subset of the hydrogen molecule. The wave numbers $\tilde{\nu}_{\mathrm{ML}}$ are computed from the predicted Hessian matrix elements. The wave numbers $\tilde{\nu}_{\mathrm{ML}}$ are plotted against GFN2–xTB calculated wave numbers $\tilde{\nu}_{\mathrm{xTB}}$. This is done for 5 random states and is shown in Fig. 4.3. The training set is 75 % of the H_2 structures, while the test set is 25 % of the H_2 structures. The black line is the diagonal, on which the data points would appear for a perfect prediction. For further discussion it is interesting to point out, that the wave number increases with the decrease of the bond length. In the range of 1800 $\mathrm{cm}^{-1} < \tilde{\nu}_{\mathrm{xTB}} < 7500\ \mathrm{cm}^{-1}$ there are small deviations from the diagonal. Therefore, with respect to the small data set of eleven structures, the model is capable to predict without significant errors.

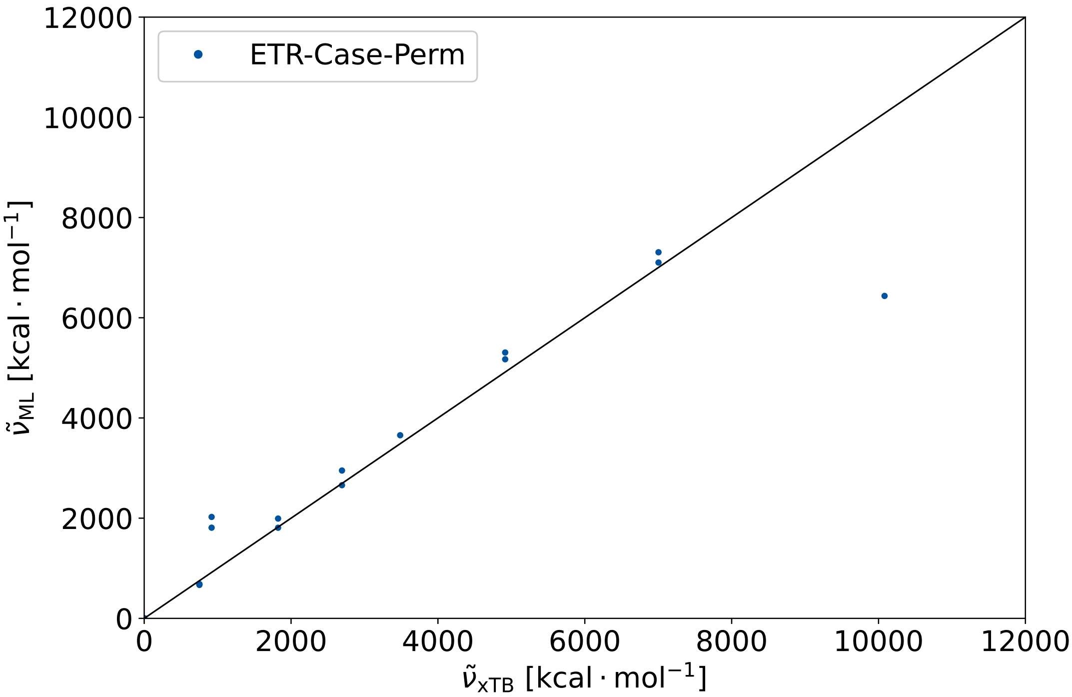


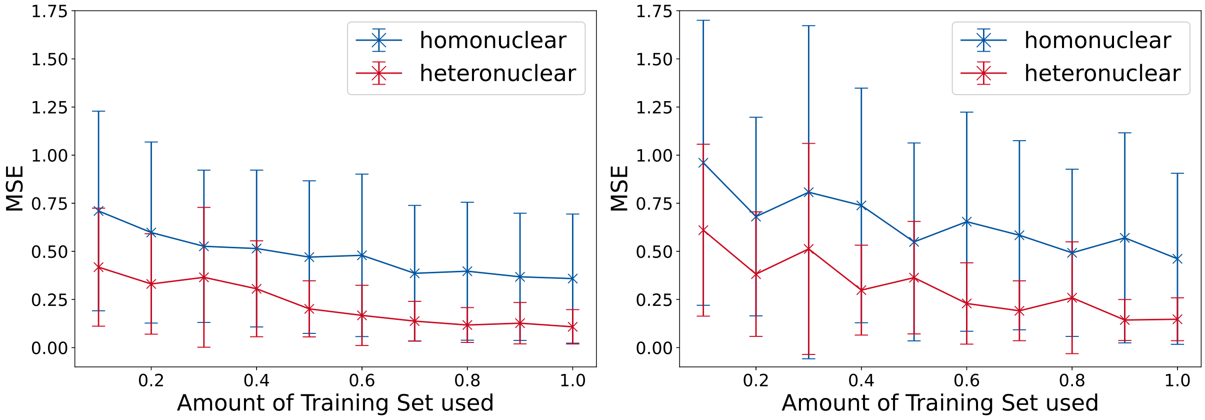
Figure 4.3: The wave number, predicted via the ML model $\tilde{\nu}_{\mathrm{ML}}$ is plotted against the wave number calculated via GFN2–xTB $\tilde{\nu}_{\mathrm{xTB}}$. The ETR–Case–Perm model is used for the prediction. The sample set is H_2 and the training set is 75 % of the whole set. The predictions are done for 5 random states.

The deviations are more significant for $\tilde{\nu}_{\mathrm{xTB}} < 1800\ \mathrm{cm}^{-1}$. For $\tilde{\nu}_{\mathrm{xTB}} > 7500\ \mathrm{cm}^{-1}$, there is an outlier. These predictions are done for cases, where the training set does not include molecules, which have a lower or higher distance, respectively. Hence, it can be assumed, that the error is due to extrapolation of the ML model. This assumption is underlined by the fact, that the predicted values correspond approximately to the next higher or lower value. For example, the next lowest value in the case of the prediction at $\tilde{\nu}_{\mathrm{ML}} \approx 1 \cdot 10^5\ \mathrm{cm}^{-1}$ is $\tilde{\nu}_{\mathrm{xTB}} \approx 7 \cdot 10^4\ \mathrm{cm}^{-1}$. This in the same order of magnitude as the predicted value $\tilde{\nu}_{\mathrm{ML}} \approx 6.4 \cdot 10^4\ \mathrm{cm}^{-1}$ at $\tilde{\nu}_{\mathrm{xTB}} \approx 1 \cdot 10^5\ \mathrm{cm}^{-1}$. Especially, this large deviation might be due to E_{rep}. The PCC showed the importance of E_{rep}, due to its high correlation.

Furthermore, E_{rep} increases significantly at low distances. Hence, the deviation due to extrapolation is significantly larger than compared to the errors at lower wave numbers (higher distances). This matches to the expected behavior for decision tree based algorithms (cf. Fig. 2.2).
This system can be seen as a proof of concept for the model and demonstrates that the extrapolation cannot be achieved.

4.4 Comparison of RFR and ETR

For the comparison of the ML algorithms, the RFR–Gen-Perm and ETR–Gen–Perm models are used. These models are compared with learning curves. Learning curves show the dependency of a model on the increase of training set used. The performance is quantified by the MSE between the predicted Hessian and the Hessian computed via GFN2–xTB. The MSE is calculated for ten different random states and the mean and standard deviation of these MSE is calculated.
It is expected, that the performance of the ML model should increase with the training set size. This can be interpreted as a statistical learning behavior. To verify this behavior, the training set is increased successively. The full training set is 75 % of the full data set. Thus, the test set is 25 % of the data set (the *Amount of Training set used* $= 1$ corresponds to 75 % of the complete data set). While keeping the test set constant, the amount of training set is changed successively from 0.1 to 1.0 of the data set.
Fig 4.4 shows the learning curves for ETR (on the left) and RFR (on the right). The error bars indicate the standard deviation. The homonuclear model is shown in blue and the heteronuclear in red.



(a) Extra Tree Regression

(b) Random Forest Regression

Figure 4.4: Learning curves for the predicted Hessian elements $\frac{\partial^2 E}{\partial a \partial b}$ are shown. On the left is the learning plot of the ETR (a) and on the right the plot of the RFR (b) shown. The full training set is 75 % of the sample set. The full sample set, excluding the H_3 system, is used. The mean of ten random states are used for the MSE. The plots are done for the homonuclear model (blue) and heteronuclear model (red).

In the case of the ETR, the MSE decreases with an increase of the amount of training set used for the homonuclear (blue) and heteronuclear (red) model. This behavior is similar in the case of the RFR. In comparison to the ETR, the RFR shows a higher standard deviation for the data points. Thus, the dependence on the random state is higher for the RFR than the ETR. Furthermore, the ETR has, especially for the homonuclear model (blue), lower MSE values for any amount of training set. For the Amount of Training Set used ≥ 0.9, the RFR and ETR show a similar performance for the heteronuclear model (red). All in all, the ETR shows a lower MSE for the whole investigated range. The features are based on physical properties and the ML model should in principle learn from these properties. This means, the more accurate the features are, the lower should the MSE be. This is tested by perturbing the features with a noise. For this, the features are scattered with a random factor. The intensity of the scattering is amplified by the factor σ. It is expected, that a model that learns from the features performs worse, if the scattering of the data points increases. The performance of the model ETR–Multi–Perm is tested for this and is shown in Fig 4.5. The full sample set, excluding the H_3 set is used for training. While the test set is the H_3 set. The mean of the MSE for ten random states are used and its standard deviation is shown. The plot is shown for the homonuclear model (blue) and the heteronuclear model (red).

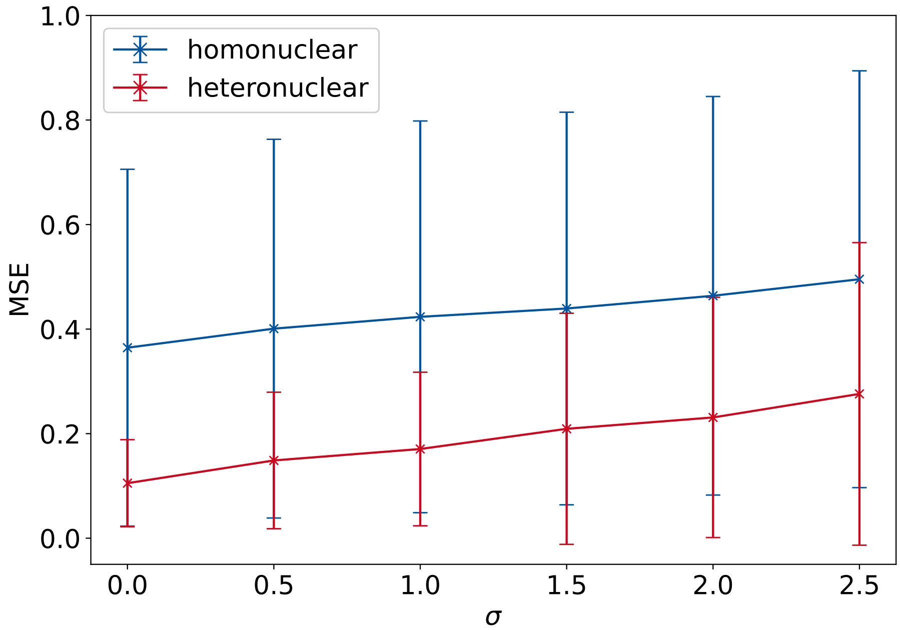


Figure 4.5: Robustness test for predicting Hessian matrix elements $\frac{\partial^2 E}{\partial a \partial b}$ for a variation of the factor σ. The full sample set, excluding the H_3 system, is used for the ML models. The mean of ten random states are used for the MSE. The plots are done for the homonuclear model (blue) and heteronuclear model (red).

The prediction is done for ten different random states. The MSE between the predicted Hessian elements and those calculated using GFN2–xTB is investigated. For the investigated range of $0 \leq \sigma \leq 2.5$ the MSE increases monotonically. Hence, this shows, that the more precise the model is, the more accurate the prediction can be.
In this section a comparison of the RFR and ETR is done. The ETR shows overall a better performance than the RFR. Furthermore, it could be demonstrated, that the ML model learns prediction correlates with the precision of the features. Hence, the Hessian matrix element prediction is not based on a random correlation with the features, but on physical information.

4.5 Comparison of Permutation Model and Indexed Model

Beside the different algorithms for the construction of the trees, the models with the variations of Gen/Case and Idx/Perm are compared. Here, the models ETR–Gen–Perm, ETR–Gen–Idx, ETR–Case–Perm and ETR–Case–Idx are investigated. The quantities for quantifying their performance are the Hessian elements and the wave numbers of the harmonic vibration. The difference between the ML predicted and GFN2–xTB computed quantities are calculated. From these differences the standard deviation and the mean are calculated, and a Gaussian distribution is assumed. At first the direct prediction of the Hessian elements is investigated, and deviation distribution is shown in Fig 4.6.

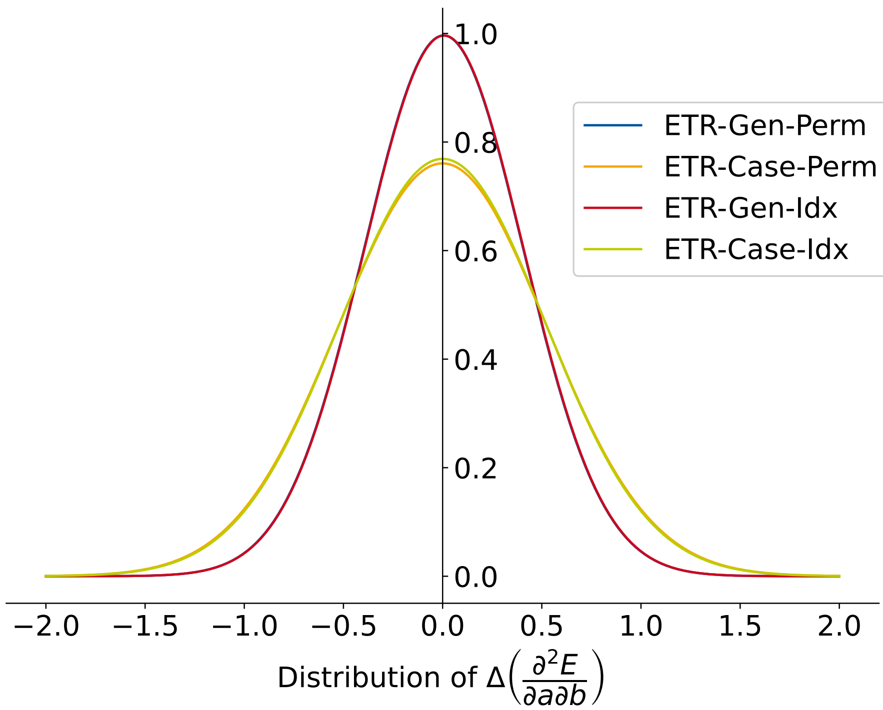
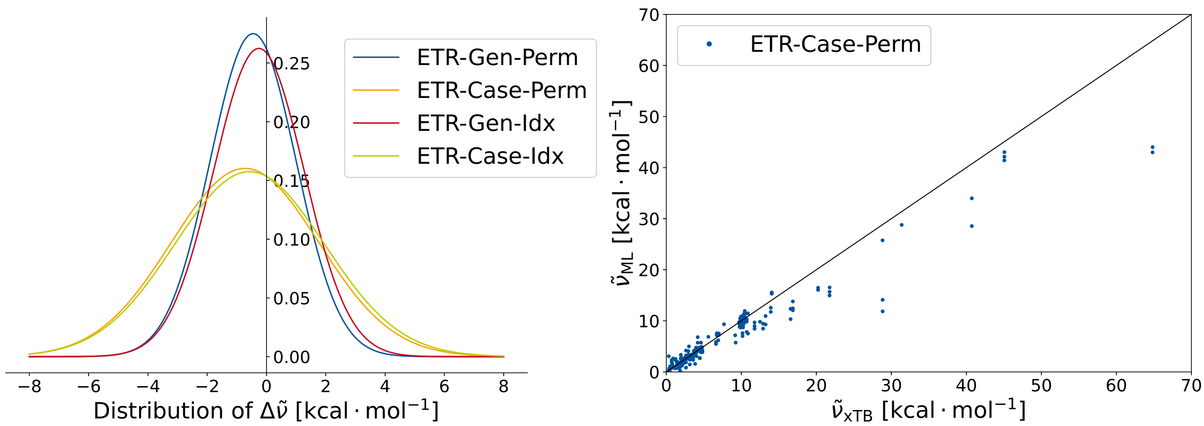


Figure 4.6: Distribution of $\Delta\left(\frac{\partial^2 E}{\partial a \partial b}\right)$ for different models. The ETR is used in every model. The distribution is assumed to be Gaussian. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E}{\partial a \partial b}\right)$ are computed.

Comparing the ETR–Case–Perm and ETR–Case–Idx, and also ETR–Gen–Perm and ETR–Gen–Idx shows, that both approaches for the labeling of the features do not differ significantly. This shows, on the one hand, the strength of the decision tree algorithms. They cannot overfit and seem to be capable of describing this complex system with a simple differentiation by indexes. The permuted model shows, on the other hand, a similar performance like the indexed model, but may be more useful for other ML alogrithms.

The ETR–Case–Perm shows larger deviations in comparison to the ETR–Gen–Perm model. This is in agreement with the comparison of the ETR–Case–Idx and ETR–Gen–Idx models. Though the deviation is more significant, the means of the models do not differ significantly. The data set for the case–specific models is smaller than for the general models, this might contribute to larger deviation. The probability, that the ML model has to predict a system by extrapolation is higher with the smaller data sets.

The wave numbers $\tilde{\nu}$ of the harmonic frequencies of the molecules are calculated as described in Sec. 3.3. The difference between the ML predicted $\tilde{\nu}_{\mathrm{ML}}$ and GFN2–xTB computed $\tilde{\nu}_{\mathrm{xTB}}$ is calculated. A Gaussian distribution is assumed. This is plotted in Fig. 4.7 for the four different varieties of the model. Furthermore, the plot of $\tilde{\nu}_{\mathrm{ML}}$ vs. $\tilde{\nu}_{\mathrm{xTB}}$ is shown.



(a) Gaussian error distribution

(b) Plot of $\tilde{\nu}_{\mathrm{ML}}$ vs. $\tilde{\nu}_{\mathrm{xTB}}$

Figure 4.7: On the left is the distribution of $\Delta\tilde{\nu}$ for different models. In every model, the ETR is used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of $\Delta\tilde{\nu}$ are computed. On the right $\tilde{\nu}_{\mathrm{ML}}$ is plotted against $\tilde{\nu}_{\mathrm{xTB}}$. The wave numbers are computed from the predicted Hessian matrix elements with the ETR–Gen–Perm model.

The relative distributions of the four variations of the models are similar to the ones of the Hessian matrix elements (cf. Fig. 4.6). The ETR–Gen–Perm and ETR–Gen–Idx show a better performance than the ETR–Case–Perm and the ETR–Case–Idx. The mean is shifted to negative $\Delta\tilde{\nu}$ values. In contrast to the deviations of the Hessian matrix elements, the Idx models show a marginally better performance than the Perm model.

The deviations of several $\mathrm{kcal} \cdot \mathrm{mol}^{-1}$ are rather strong. These strong deviations might be due to high extrapolation errors, which lead to high standard distribution. For example at $\tilde{\nu}_{\mathrm{xTB}} \approx 65\mathrm{kcal} \cdot \mathrm{mol}^{-1}$, the deviation might be due to models extrapolating values. Hence, the difference of about $\Delta\tilde{\nu} \approx 20\mathrm{kcal} \cdot \mathrm{mol}^{-1}$ has a strong influence on the distribution, although ETR and RFR is not customized for extrapolation problems. A contrary effect to the extrapolation errors are the wave numbers, which correspond to rotation and translation. These wave numbers are approximately zero due to the projection and have large contribution to amount of wave numbers computed. Therefore, they narrow the standard deviation, and shift the mean to zero. Further investigations are necessary, to see whether the extrapolation or the wave numbers, corresponding to rotation and translation, have a higher impact on the mean and standard deviation.

4.6 Benchmarking with the H_3 System

Especially, the prediction of Hessian matrices as a transfer application is of interest. Hence, the ETR-Gen-Perm and ETR–Gen–Idx models are tested on the performance on a new, for the model unknown, system. Furthermore, the chosen H_3 system shows a symmetric behavior, which should also be described by the ML models. The H_3 system is constructed of 3 hydrogen atoms, where the outer hydrogen atoms are fixed, while the inner hydrogen atom is shifted from one hydrogen to another. H1 is fixed at $x(\mathrm{H1}) = 0.0$ Å and H3 at $x(\mathrm{H1}) = 4.0$ Å. For each position of the hydrogen atom, the force constants of the systems are predicted via the ML model. The force constants are also calculated via GFN2–xTB.

Fig. 4.8 shows the force constants calculated by GFN2–xTB and predicted via the ETR–Gen–Idx model. The ML model shows small deviation in respect to the GFN2–xTB computed k. In general the ML overestimates the force constants k. The symmetry is overall preserved. Though, for some cases the symmetry is broken. For example, the symmetry is broken in the case of $x(\mathrm{H2}) \approx 1.84$ Å and $x(\mathrm{H2}) \approx 2.15$ Å.

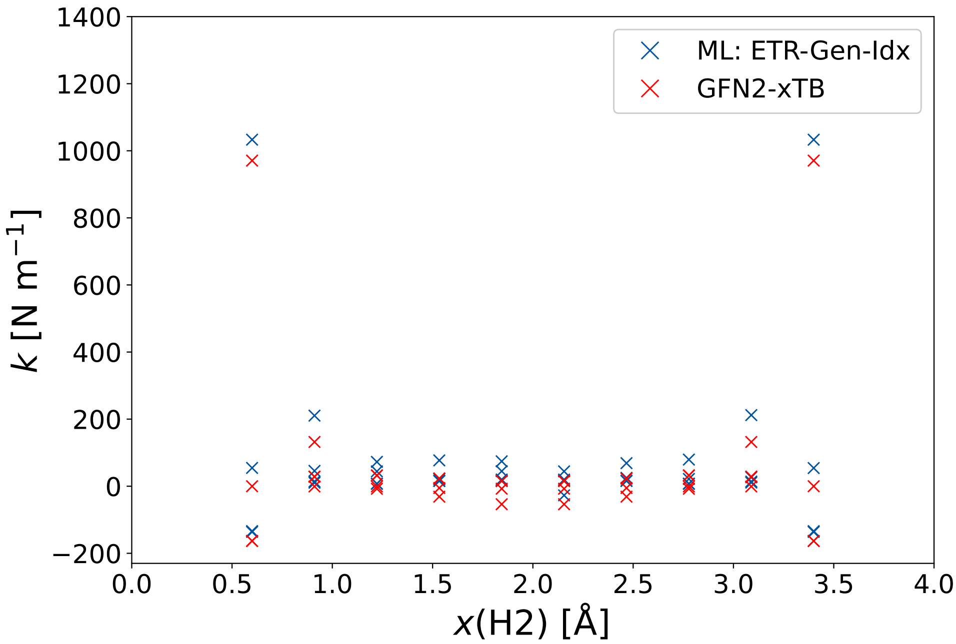


Figure 4.8: Force constants of the H_3 system as a function of the distance between H1 and H2. H1 is located at $x(\mathrm{H1}) = 0.0$ Å and H3 at $x(\mathrm{H3}) = 4.0$ Å. The force constants are shown for the GFN2–xTB calculation (red), and the predicted values (blue). The full sample set, excluding the H_3 system, is used for the ML model. The ML model ETR–Gen–Idx is used for the prediction.

Fig. 4.9 shows the prediction of the force constants via the ETR–Gen–Perm model. The GFN2–xTB computed k is shown in red, the ML predicted is blue. The full sample set is used as the training set, but the H_3–structures. Similar to the ETR–Gen–Idx model, the symmetry is in general reproduced. Nevertheless, for example the force constants at $x(\mathrm{H2}) \approx 0.91$ Å and $x(\mathrm{H2}) \approx 3.08$ Å show deviations.

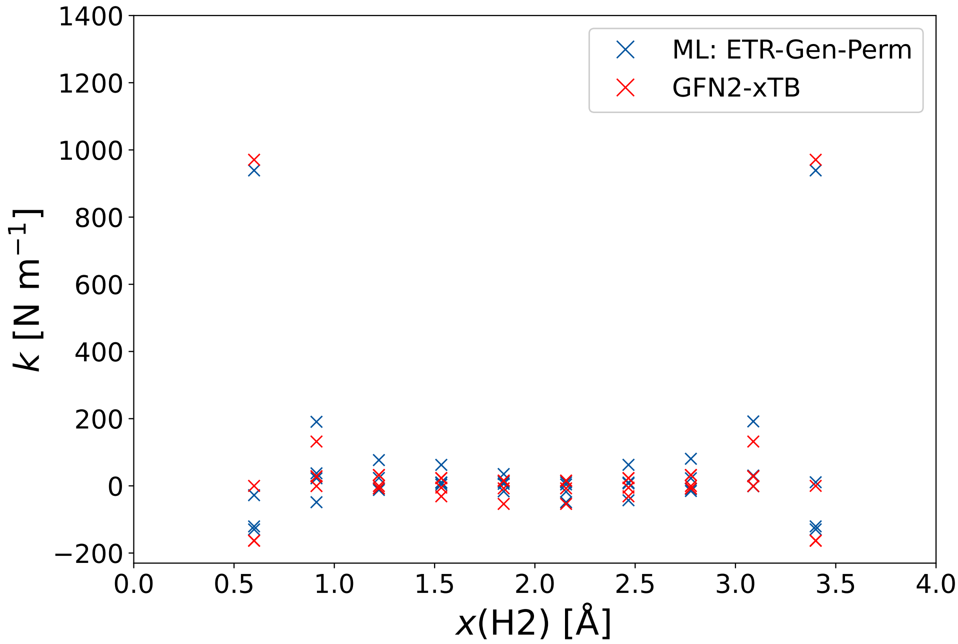


Figure 4.9: Force constants of the H_3 system as a function of the distance between H1 and H2. H1 is located at $x(\mathrm{H1}) = 0.0$ Å and H3 at $x(\mathrm{H3}) = 4.0$ Å The force constants are shown for the GFN2–xTB calculation (red), and the predicted values (blue). The full sample set, excluding the H_3 system, is used for the ML model. The ML model ETR–Gen–Perm is used for the prediction.

A prediction of the force constants k with overall small deviations is achieved, though there are still errors in the symmetry. Further investigations on the models are necessary, to locate the reason for the break of symmetry. At first a decoupled investigation on the homonuclear and heteronuclear model is of interest, as their features differ. Especially the processing of the GFN2–xTB quantities for the heteronuclear model might be an error source. Furthermore, the vector and matrix features depend on the rotation of the CS and therefore also could be responsible for errors in the symmetry.

5 Conclusion & Outlook

The goal of this research project was the prediction of Hessian matrices by a ML model. The ML model is based on atomic features and the prediction of the elements of the Hessian matrices is done atom pair wise. The model is benchmarked by a small data set of 8 molecules with 103 different structures. The computations of the Hessian matrices and features for these systems is done with the GFN2–xTB method. These computed quantities were used for the training of the ML models and also as the reference.
At first, the general learning of the model was investigated. A statistical learning behavior of the ML models RFR–Gen–Perm and ETR–Gen–Perm could be observed. The ETR–Gen–Perm model showed a better performance. Furthermore, a physical correlation could be demonstrated for the ETR–Gen–Perm model.
Moreover, four variations of the model are investigated and compared to another. Here, a similar behavior for different labeling approaches (Indexed and Permuted) was observed. The general models show a better prediction than the case–specific models. The performance was evaluated by computing the prediction error of the Hessian matrix elements and the calculated wave numbers. In addition, the set is used for transfer learning.
The data set of the new H_3 system shows an almost symmetric behavior for the force constants, which could, with some minor errors, be depicted by the ML models.
The next necessary approach for this project, is the improvement of the models in their symmetric behavior. The adjustment to a full symmetric behavior should be mandatory. As the correlation of the features is low, investigations towards higher correlating features might increase the performance of the ML models. Here, new processing of the GFN2–xTB quantities could be used, which achieve a linear correlation between the features and target. In addition, a set of 103 structures is small for a ML model. Hence, the date set size should be increased and tests on benchmark data sets should be done for a comparison with literature. Instead of predicting Hessian matrix elements directly on GFN2–xTB level, the model could be adapted to a ΔML model. For this model the features from GFN2–xTB could be used to predict the difference between the Hessian matrix elements on GFN2–xTB and a higher level metohd.

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7 Appendix

$$CN_A = \sum_{B \neq A}^{N_{\text{atoms}}} \frac{1}{1 + \exp\left(\frac{-16a \cdot 4(R_{A,\text{cov}} + R_{B,\text{cov}})}{3R_{AB} - 1}\right)} \tag{7.0.1a}$$

$$CN^{\text{ext}} = \sum_{B \neq A}^{N_{\text{atoms}}} \frac{CN_B}{f_{\text{damp}}(R_A, R_B)} \tag{7.0.1b}$$

$$q_A = Zg_A - \sum_{\mu \in A}^{N_{\text{BF}}} \sum_{\nu}^{N_{\text{BF}}} P_{\mu\nu} S_{\mu\nu} \tag{7.0.1c}$$

$$q_A^{\text{ext}} = q_A + \sum_{B \neq A}^{N_{\text{atoms}}} \frac{q_B}{f_{\text{damp}}} \tag{7.0.1d}$$

$$\mu_A^{\alpha} = \sum_{l \in A} \sum_{\kappa \in l \in A} \sum_{\lambda} P_{\kappa\lambda} \left(\alpha_A S_{\lambda\kappa} - D_{\lambda\kappa}^{\alpha}\right) \tag{7.0.1e}$$

$$\mu_A^{\alpha,\text{ext}} = \mu_A^{\alpha} + \sum_{B \neq A} \frac{\mu_B^{\alpha} - \alpha_{AB} q_B}{f_{\text{damp}} \cdot (CN_B + 1)} \tag{7.0.1f}$$

$$\text{with}: \ \alpha_{AB} = \alpha_A - \alpha_B \tag{7.0.1g}$$

$$\mu_{A,p}^{\alpha,\text{ext}} = \mu_A^{\alpha} + \sum_{B \neq A} \frac{\mu_B^{\alpha} + \alpha_{AB} p_B}{f_{\text{damp}}(CN_B + 1)} \tag{7.0.1h}$$

$$\theta_A^{\alpha\beta} = \sum_{l \in A} \frac{3}{2} \Theta_l^{\alpha\beta} - \frac{\delta^{\alpha\beta}}{2} (\Theta_l^{xx} + \Theta_l^{yy} + \Theta_l^{zz}) \tag{7.0.1i}$$

$$\text{with}: \ \Theta_l^{\alpha\beta} = \sum_{\kappa \in l \in A} \sum_{A} P_{\kappa\lambda} \left(\alpha_A D_{\lambda\kappa}^{\beta} + \beta_A D_{\lambda\kappa}^{\alpha} - \alpha_A \beta_A S_{\lambda\kappa} - Q_{\lambda\kappa}^{\alpha\beta}\right)$$

$$\text{and}: \ \delta^{\alpha\beta} = \begin{cases} 1 \text{ if} : \alpha = \beta \\ 0 \text{ if} : \alpha \neq \beta \end{cases}$$

$$\theta_A = \begin{pmatrix} \theta_A^{xx} & \theta_A^{xy} & \theta_A^{xz} \\ \theta_A^{xy} & \theta_A^{yy} & \theta_A^{yz} \\ \theta_A^{xz} & \theta_A^{yz} & \theta_A^{yz} \end{pmatrix} \times \mathbf{1} \tag{7.0.1j}$$

$$\theta_A^{\text{ext},\alpha\beta} = \theta_A^{\alpha\beta} + \frac{\theta_B^{\alpha\beta} + \delta\theta_B^{\alpha\beta}}{f_{\text{damp}}(CN_B + 1)} \tag{7.0.1k}$$

$$\theta_A^{\text{ext}} = \begin{pmatrix} \theta_A^{\text{ext},xx} & \theta_A^{\text{ext},xy} & \theta_A^{\text{ext},xz} \\ \theta_A^{\text{ext},xy} & \theta_A^{\text{ext},yy} & \theta_A^{\text{ext},yz} \\ \theta_A^{\text{ext},xz} & \theta_A^{\text{ext},yz} & \theta_A^{\text{ext},yz} \end{pmatrix} \times \mathbf{1} \tag{7.0.1l}$$

$$E_{A,\mathrm{EHT}} = \sum_{k\in A}\sum_{k} P_{k\lambda} H_{k\lambda} \tag{7.0.1m}$$

$$E_{A,\mathrm{rep}} = \sum_{B\neq A}^{M} \frac{1}{2}\frac{Y_A^{\mathrm{eff}} Y_B^{\mathrm{eff}}}{R_{AB}} \cdot \exp\left(\alpha_A \alpha_B\right)^{0.5} R_{AB}^{k_{\mathrm{rep}}} \tag{7.0.1n}$$

$$E_{A,\mathrm{disp}}^{(2)} = \sum_{A\neq B}\sum_{n=6,8} -s_n \frac{1}{2}\frac{C_n^{AB}(q_a, CN_{\mathrm{cov}}^{A}, q_b, CN_{\mathrm{cov}}^{B})}{R_{AB}^{n}} f_n^{\mathrm{damp,BJ}}(R_{AB}) \tag{7.0.1o}$$

$$\begin{aligned} E_{A,\mathrm{disp}}^{(3)} &= \sum_{B\neq A}\sum_{C\neq B\neq A} -s_9 \frac{1}{3}\frac{\left(3\cos(\theta_{ABC})\cos(\theta_{BCA}\cos(\theta_{CBA})+1)\right)}{(R_{AB}R_{AC}R_{BC})^3} \\ &\quad \cdot C_9^{ABC}(CN_{\mathrm{cov}}^{A}, CN_{\mathrm{cov}}^{B}, CN_{\mathrm{cov}}^{C}) \cdot f_9^{\mathrm{damp,zero}}(R_{AB}, R_{AC}, R_{BC}) \end{aligned} \tag{7.0.1p}$$

$$\begin{aligned} E_{A,\mathrm{AES}} &= E_{A,q\mu} + E_{A,q\theta} + E_{A,\mu\mu} \\ &= \sum_{B\neq A}\frac{1}{2}\Bigg(f_3(R_{AB})\left[q_A(\boldsymbol{\mu}_B^{\mathsf{T}}\mathbf{R}_{BA}) + q_B(\boldsymbol{\mu}_A^{\mathsf{T}})\mathbf{R}_{AB}\right] \\ &\quad + f_5(R_{AB})\big[q_A \boldsymbol{R}_{BA}^{\mathsf{T}}\boldsymbol{\Theta}_B \boldsymbol{R}_{BA} + q_B \boldsymbol{R}_{BA}^{\mathsf{T}}\boldsymbol{\Theta}_A \boldsymbol{R}_{BA} \\ &\quad - 3(\boldsymbol{\mu}_A^{\mathsf{T}}\boldsymbol{R}_{AB})(\boldsymbol{\mu}_B^{\mathsf{T}}\boldsymbol{R}_{AB}) + (\boldsymbol{\mu}_A^{\mathsf{T}}\boldsymbol{\mu}_B R_{AB}^2)\big]\Bigg) \end{aligned} \tag{7.0.1q}$$

$$E_{A,\mathrm{AXS}} = \left(f_{\mathrm{XC}}^{\mu_A}|\boldsymbol{\mu}_A|^2 + f_{\mathrm{XC}}^{\Theta_A}||\boldsymbol{\Theta}_A||^2 \right) \tag{7.0.1r}$$

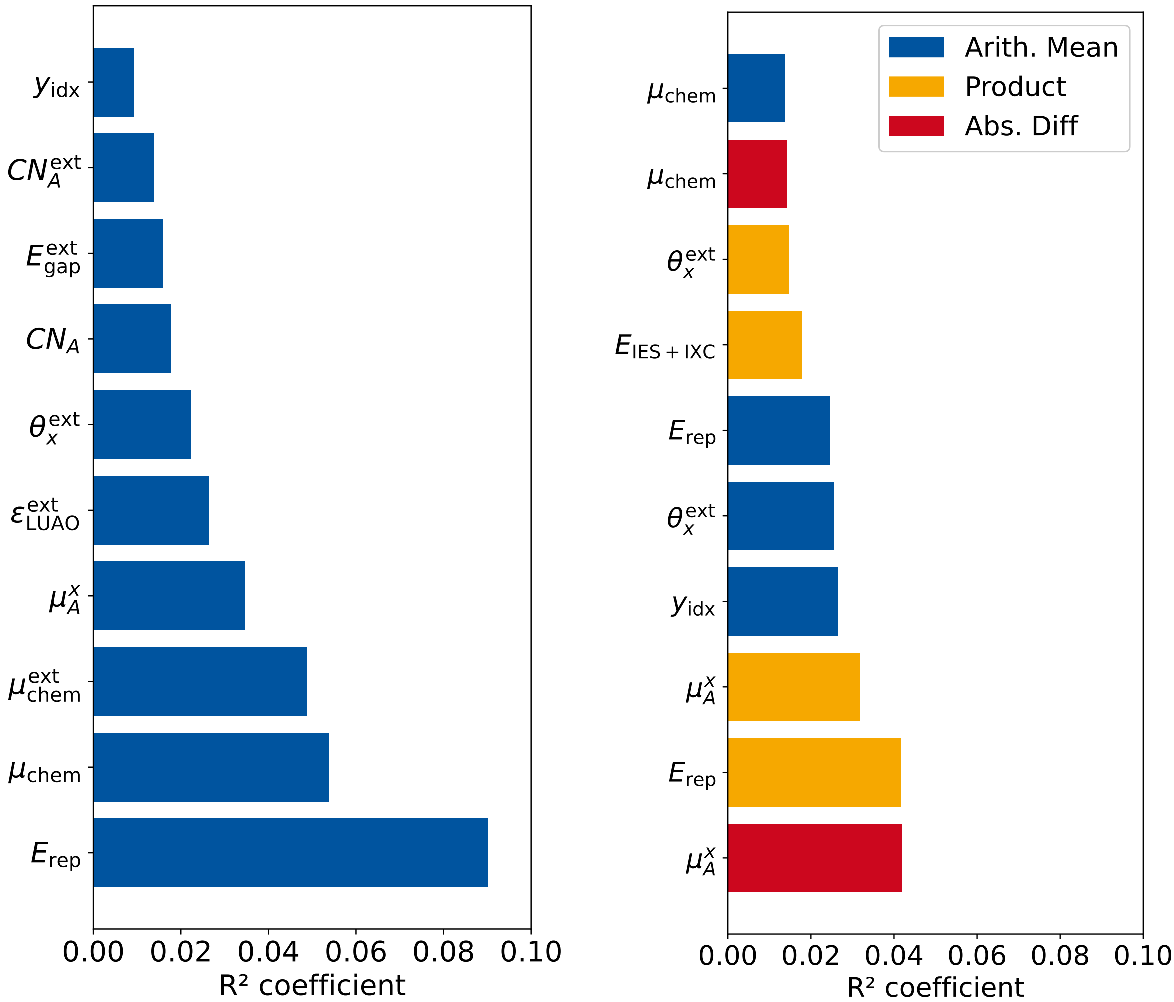
$$E_{A,\mathrm{IES+IXC}} = \frac{1}{2}\sum_{B\neq A}\sum_{l\in A}\sum_{l'\in B} q_{A,l}\, q_{B,l'}\, \gamma_{AB,ll'} + \frac{1}{3}\sum_{l\in A}\Gamma_{A,l}\, q_{A,l}^3 \tag{7.0.1s}$$

Table 7.1: Indices used for the different elements in a Hessian matrix block to differ between the elements.

Hessian element	Indexed x^{idx}	y^{idx}	z^{idx}	Permuted Index
$\frac{\partial^2 E}{\partial x_A \partial x_B}$	2	0	0	0
$\frac{\partial^2 E}{\partial x_A \partial y_B}$	1	-1	0	1
$\frac{\partial^2 E}{\partial x_A \partial z_B}$	1	0	-1	2
$\frac{\partial^2 E}{\partial y_A \partial x_B}$	-1	1	0	1
$\frac{\partial^2 E}{\partial y_A \partial y_B}$	0	2	0	0
$\frac{\partial^2 E}{\partial y_A \partial z_B}$	0	1	-1	2
$\frac{\partial^2 E}{\partial z_A \partial x_B}$	-1	0	1	2
$\frac{\partial^2 E}{\partial z_A \partial y_B}$	0	-1	1	2
$\frac{\partial^2 E}{\partial z_A \partial z_B}$	0	0	2	3

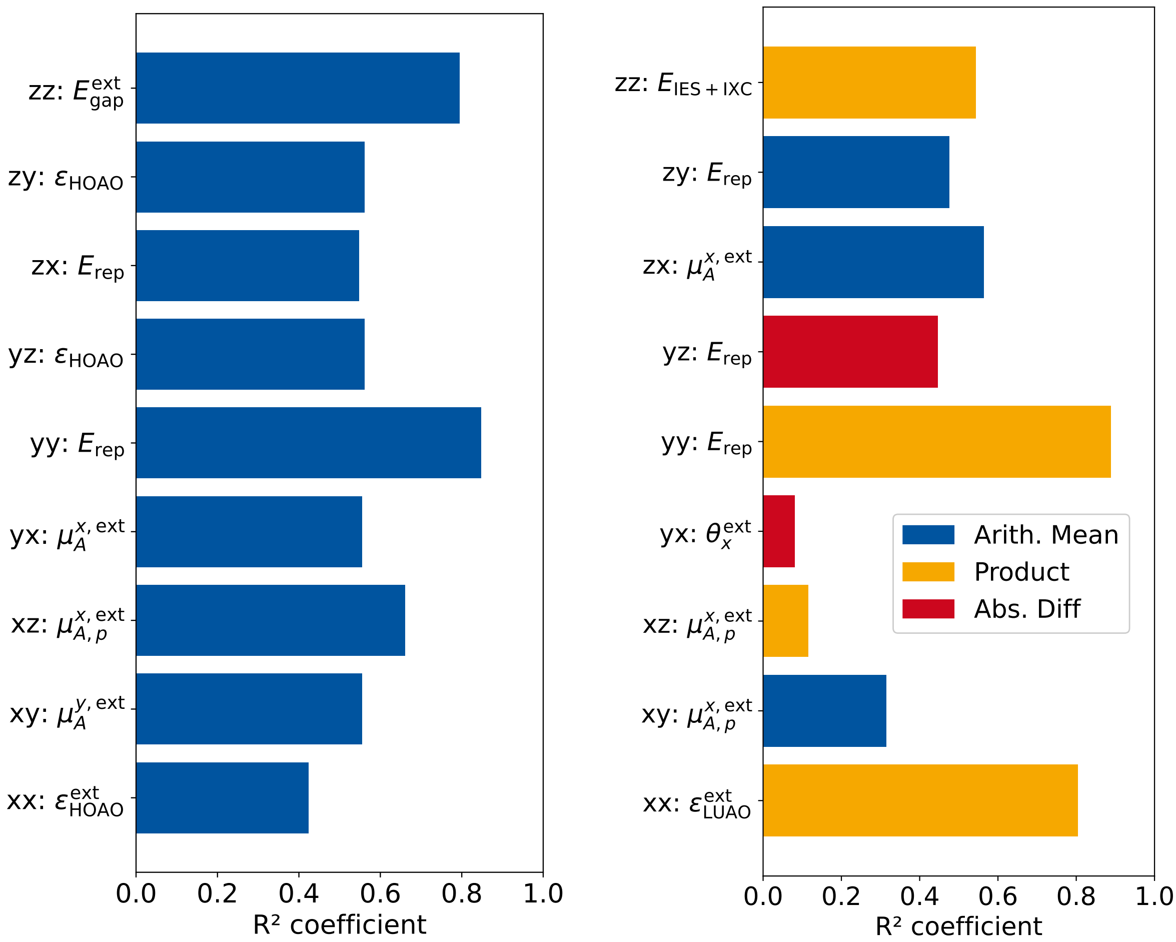
Table 7.2: Permutation of the coordinates for each Hessian element. At first the permutation of the vectors is shown. After that, the change of the elements of a matrix, for example, for the quadrupole moment is shown.

Hessian	$\frac{\partial^2 E}{\partial x_A \partial x_A}$	$\frac{\partial^2 E}{\partial x_A \partial y_A}$	$\frac{\partial^2 E}{\partial x_A \partial z_A}$	$\frac{\partial^2 E}{\partial y_A \partial x_A}$	$\frac{\partial^2 E}{\partial y_A \partial y_A}$	$\frac{\partial^2 E}{\partial y_A \partial z_A}$	$\frac{\partial^2 E}{\partial z_A \partial x_A}$	$\frac{\partial^2 E}{\partial z_A \partial y_A}$	$\frac{\partial^2 E}{\partial z_A \partial z_A}$
Vector									
q_x	q_x	q_y	q_z	q_x	q_y	q_z	q_x	q_y	q_z
q_y	q_y	q_x	q_x	q_y	q_z	q_y	q_z	q_z	q_x
q_z	q_z	$-q_z$	q_y	q_z	q_x	$-q_x$	$-q_y$	q_x	q_y
Matrix									
Q_{xx}	Q_{xx}	Q_{yy}	Q_{zz}	Q_{xx}	Q_{yy}	Q_{zz}	Q_{xx}	Q_{yy}	Q_{zz}
Q_{yy}	Q_{xy}	Q_{xx}	Q_{xx}	Q_{yy}	Q_{zz}	Q_{yy}	Q_{zz}	Q_{zz}	Q_{xx}
Q_{zz}	Q_{xz}	Q_{zz}	Q_{yy}	Q_{zz}	Q_{xx}	Q_{xx}	Q_{yy}	Q_{xx}	Q_{yy}
Q_{xy}	Q_{xy}	Q_{xy}	Q_{xz}	Q_{xy}	Q_{yz}	Q_{yz}	Q_{xz}	Q_{yz}	Q_{xz}
Q_{xz}	Q_{xz}	$-Q_{yz}$	Q_{yz}	Q_{xz}	Q_{xy}	$-Q_{xz}$	$-Q_{xy}$	Q_{xy}	Q_{yz}
Q_{yz}	Q_{yz}	$-Q_{xz}$	Q_{xy}	Q_{yz}	Q_{xz}	$-Q_{xy}$	$-Q_{yz}$	Q_{xz}	Q_{xy}



(a) homonuclear model (b) heteronuclear model

Figure 7.1: Pearson Correlation Coefficient for the features used for the homonuclear and heteronuclear model. The ten features with the highest coefficients are shown for each model. The features are constructed with the Gen–Perm model.



(a) homonuclear model

(b) heteronuclear model

Figure 7.2: Highest Pearson Correlation Coefficient specifically computed for each Hessian element for the homo– and heteronuclear model. The abbreviation ab corresponds to the elements $\frac{\partial^2 E}{\partial a_A \partial b_B}$. The features are constructed with the Gen–Perm model.

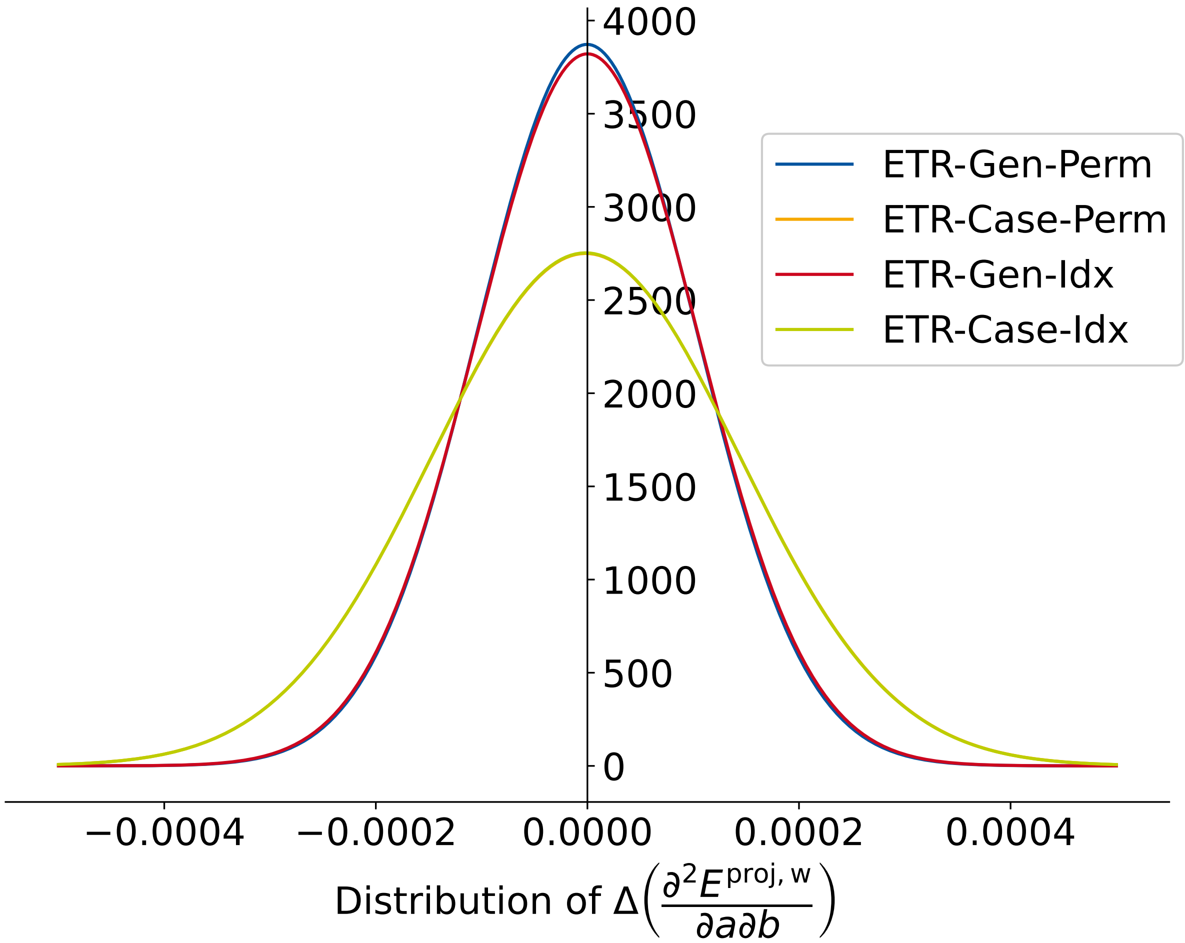


Figure 7.3: Distribution of $\Delta\left(\frac{\partial^2 E^{\text{proj,w}}}{\partial a \partial b}\right)$ for different models. In every model, the ETR was used. A Gaussian distribution is assumed. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E^{\text{proj,w}}}{\partial a \partial b}\right)$ are computed.

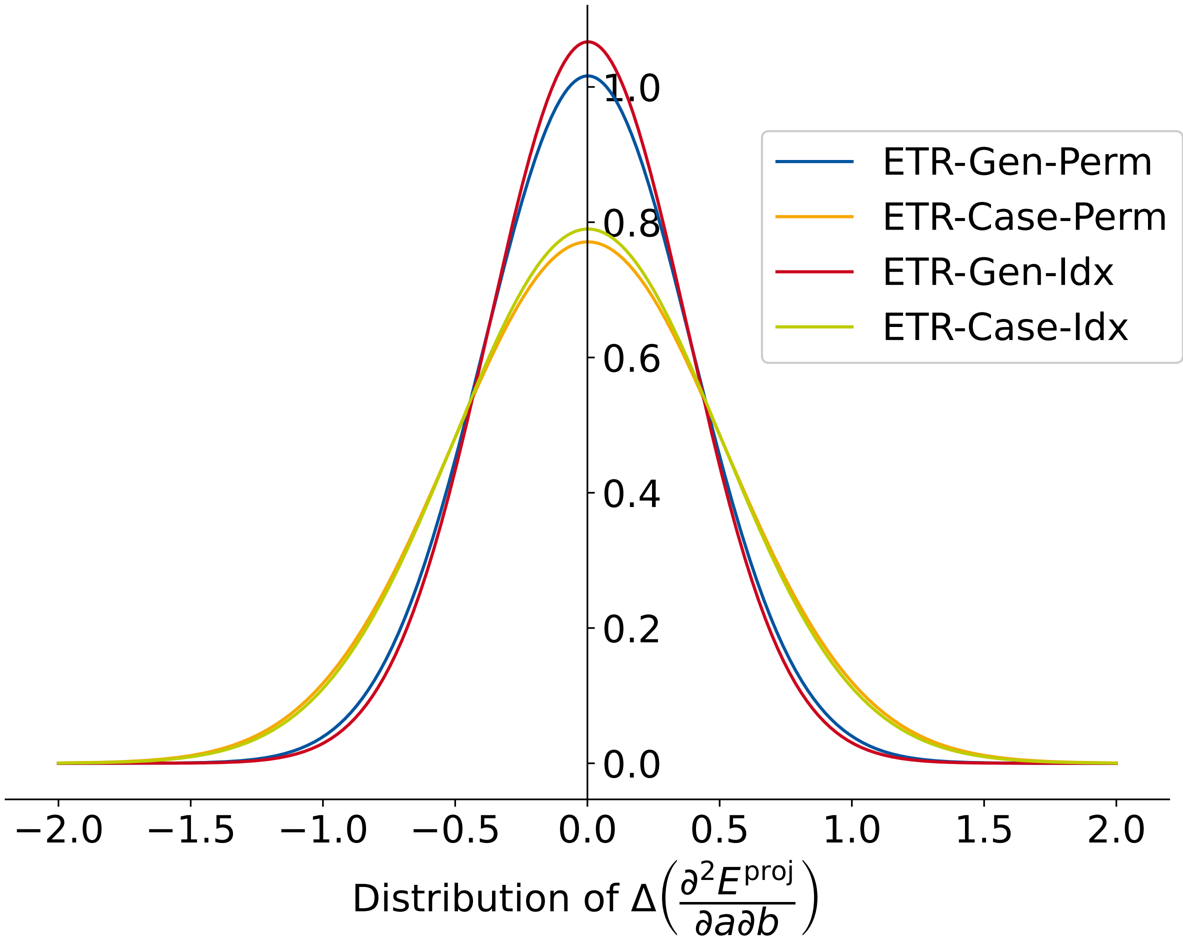


Figure 7.4: Distribution of $\Delta\left(\frac{\partial^2 E^{\text{proj}}}{\partial a \partial b}\right)$ for different models. In every model, the ETR was used. A Gaussian distribution is assumed. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E^{\text{proj}}}{\partial a \partial b}\right)$ are computed.

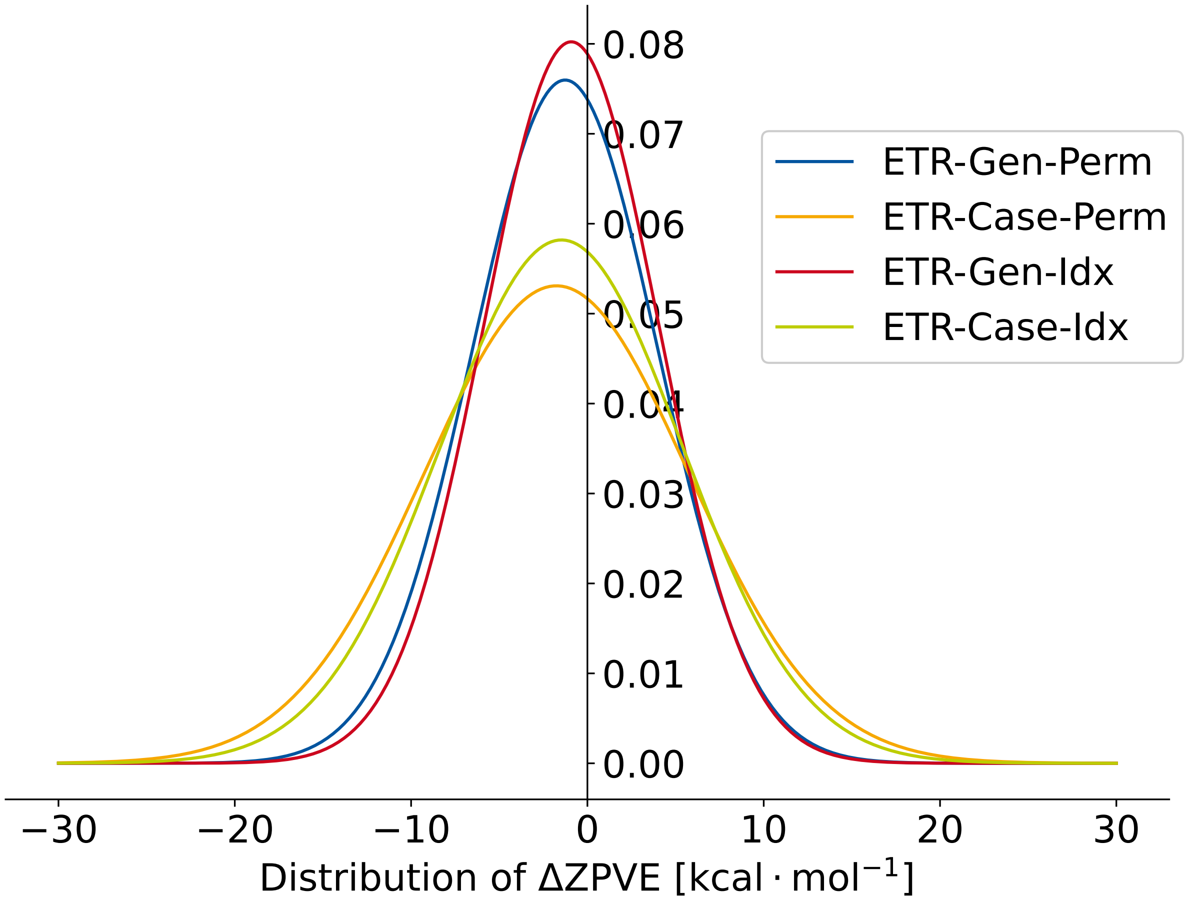


Figure 7.5: Distribution of ΔZPE for different models. In every model, the ETR was used. A Gaussian distribution is assumed. For this the median and standard deviation of ΔZPE are computed.